

# Fort Pickens State Park Aquatic Preserve

## SEACAR Water Quality Analysis

Last compiled on 05 May, 2026

### Contents

|                                                                |          |
|----------------------------------------------------------------|----------|
| <b>Indicators</b>                                              | <b>2</b> |
| Nutrients . . . . .                                            | 2        |
| Total Nitrogen - Discrete . . . . .                            | 2        |
| Total Phosphorus - Discrete . . . . .                          | 4        |
| Water Quality . . . . .                                        | 6        |
| Dissolved Oxygen - Continuous . . . . .                        | 6        |
| Dissolved Oxygen - Discrete . . . . .                          | 8        |
| Dissolved Oxygen Saturation - Continuous . . . . .             | 10       |
| Dissolved Oxygen Saturation - Discrete . . . . .               | 12       |
| Salinity - Discrete . . . . .                                  | 14       |
| Salinity - Continuous . . . . .                                | 16       |
| Water Temperature - Continuous . . . . .                       | 18       |
| Water Temperature - Discrete . . . . .                         | 20       |
| pH - Continuous . . . . .                                      | 22       |
| pH - Discrete . . . . .                                        | 24       |
| Water Clarity . . . . .                                        | 26       |
| Turbidity - Continuous . . . . .                               | 26       |
| Turbidity - Discrete . . . . .                                 | 28       |
| Total Suspended Solids - Discrete . . . . .                    | 30       |
| Chlorophyll a, Uncorrected for Pheophytin - Discrete . . . . . | 32       |
| Chlorophyll a, Corrected for Pheophytin - Discrete . . . . .   | 34       |
| Secchi Depth - Discrete . . . . .                              | 36       |
| Colored Dissolved Organic Matter - Discrete . . . . .          | 38       |

# Indicators

## Nutrients

### Total Nitrogen - Discrete

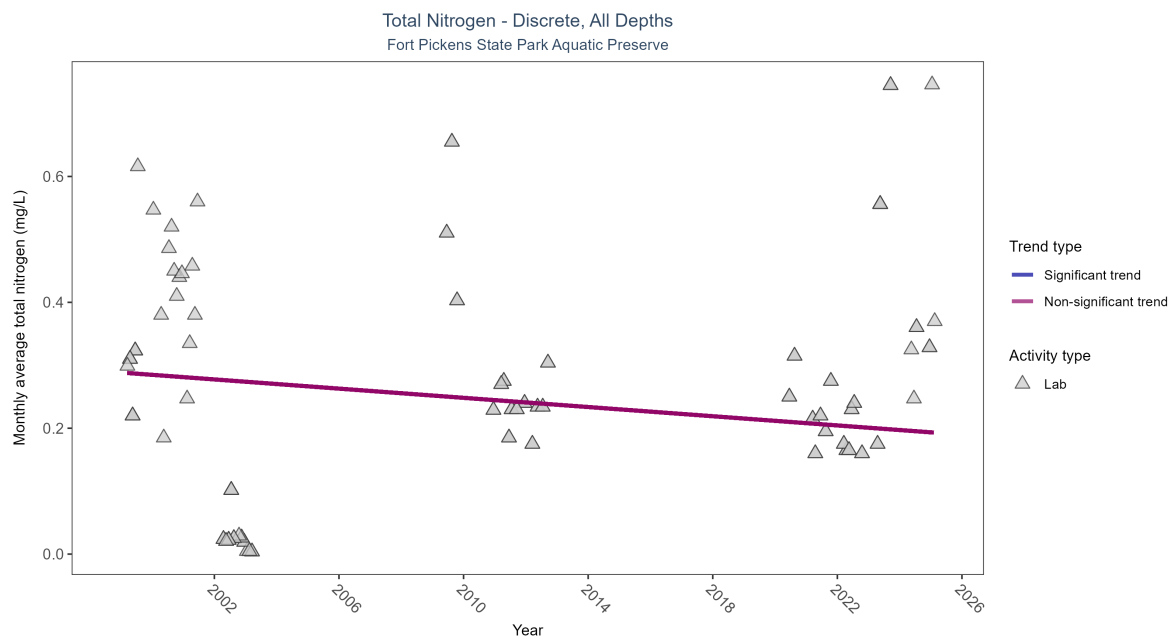


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

| Activity Type | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P      |
|---------------|----------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|--------|
| Lab           | No significant trend | 119          | 15              | 1999 - 2025      | 0.245               | -0.08647 | 0.28849       | -0.00366  | 0.3742 |

Total nitrogen showed no detectable trend between 1999 and 2025.

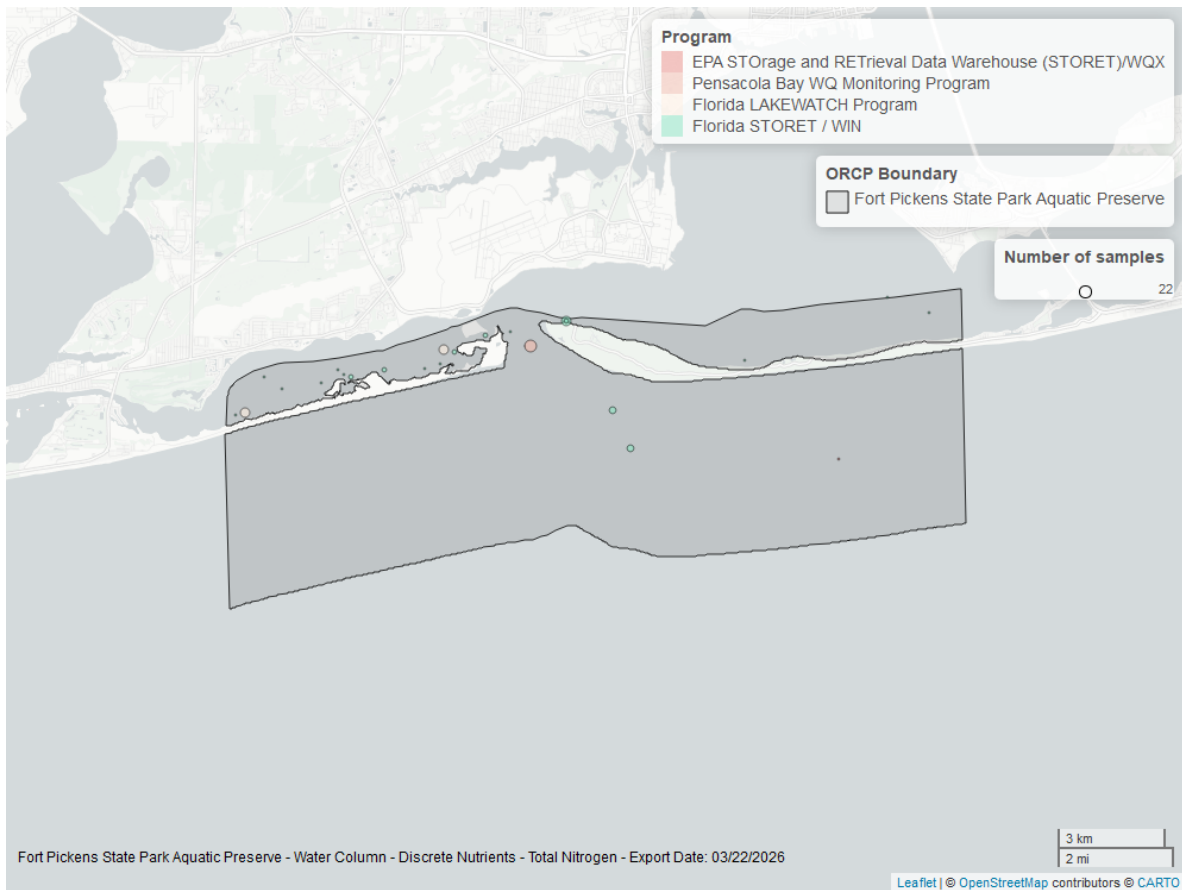


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Total Phosphorus - Discrete

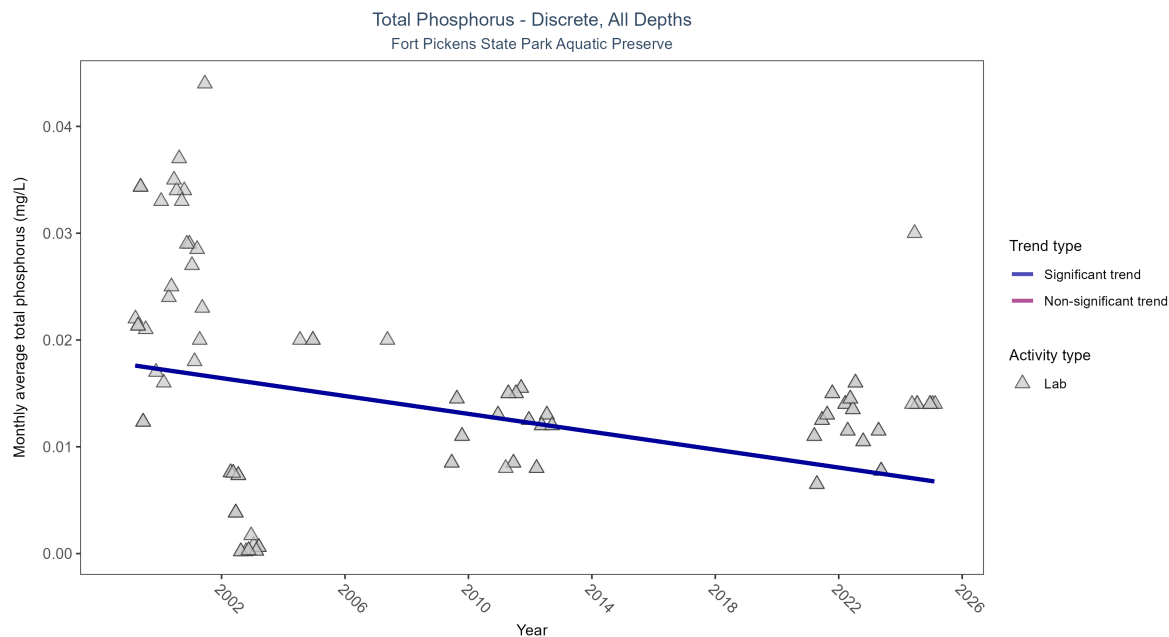


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|--------|
| Lab           | Significantly decreasing trend | 118          | 16              | 1999 - 2025      | 0.0135              | -0.32037 | 0.01769       | -0.00042  | 0.0072 |

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

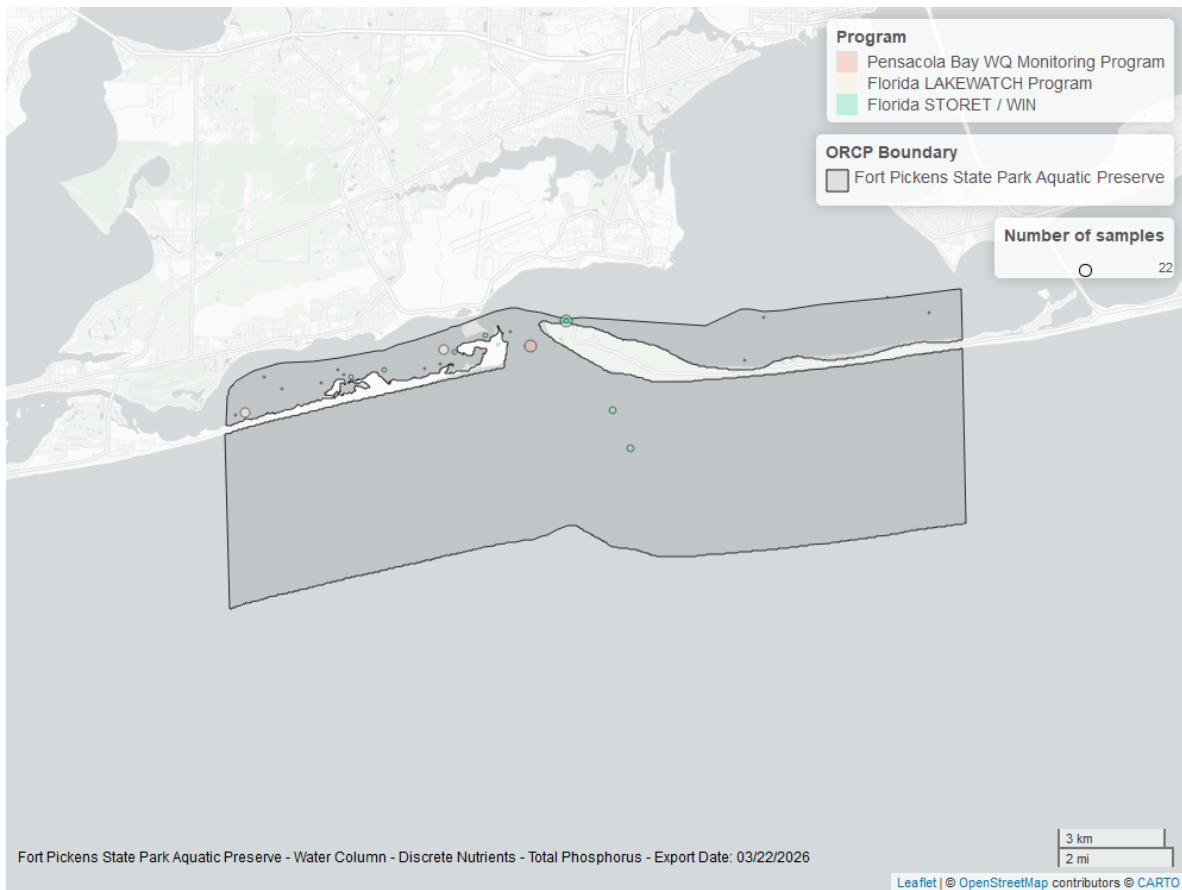


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Quality

### Dissolved Oxygen - Continuous

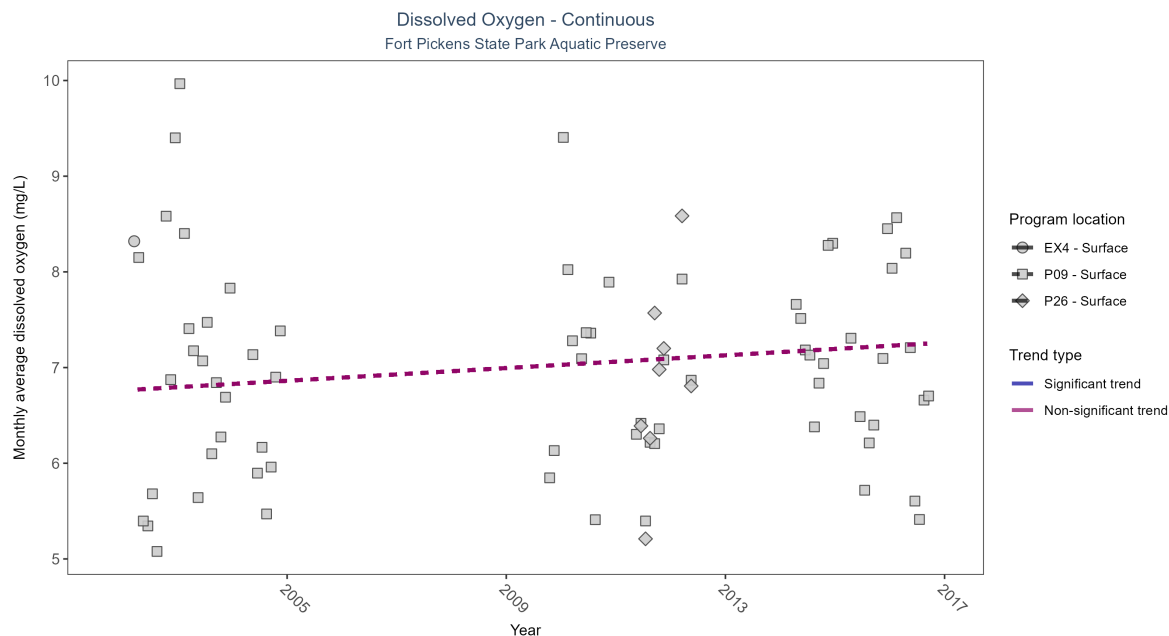


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P     |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|------|---------------|-----------|-------|
| EX4              | Insufficient data to calculate trend | 2            | 1               | 2002 - 2002      | 8.32                | -    | -             | -         | -     |
| P09              | No significant trend                 | 364          | 10              | 2002 - 2016      | 6.82                | 0.14 | 6.76          | 0.03      | 0.097 |
| P26              | Insufficient data to calculate trend | 27           | 2               | 2011 - 2012      | 6.61                | -    | -             | -         | -     |

No detectable change in monthly average dissolved oxygen was observed at one location. There was insufficient data to fit a model for two locations.

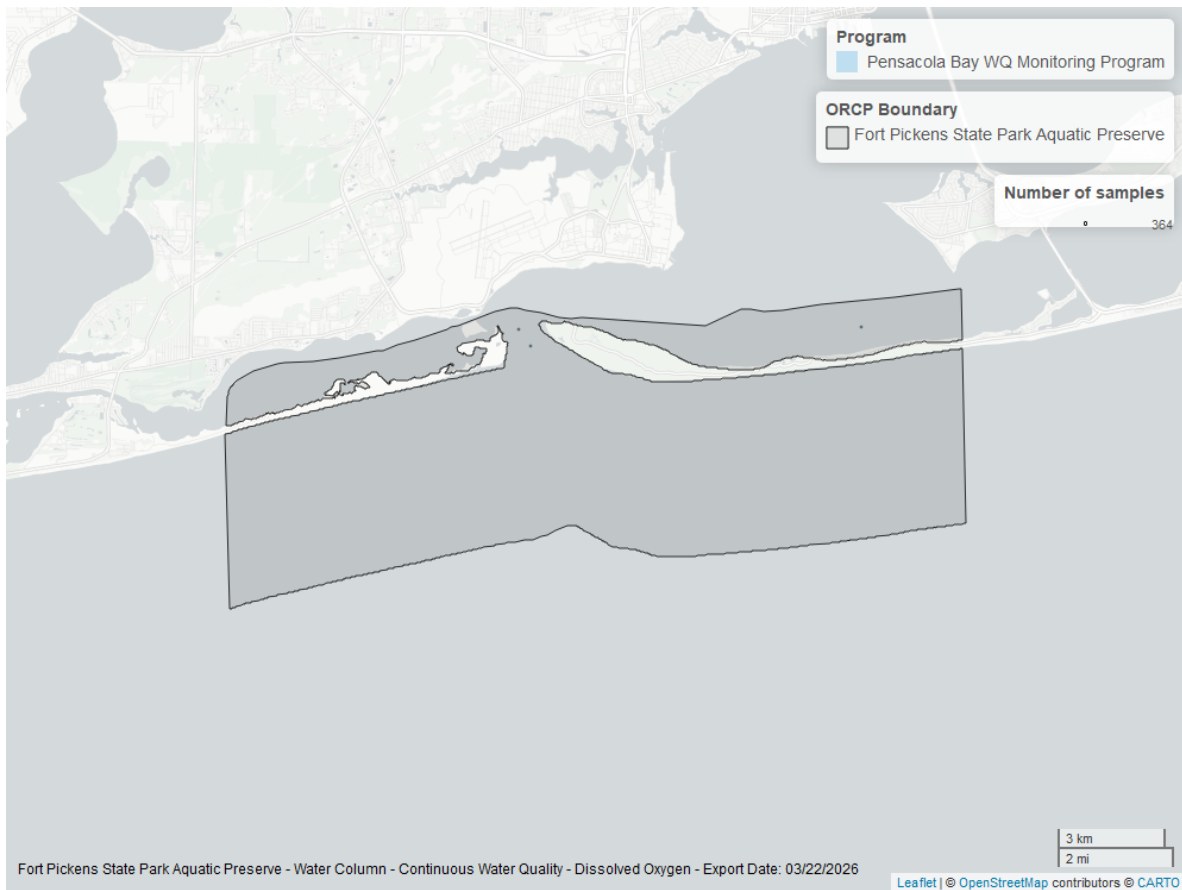


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Dissolved Oxygen - Discrete

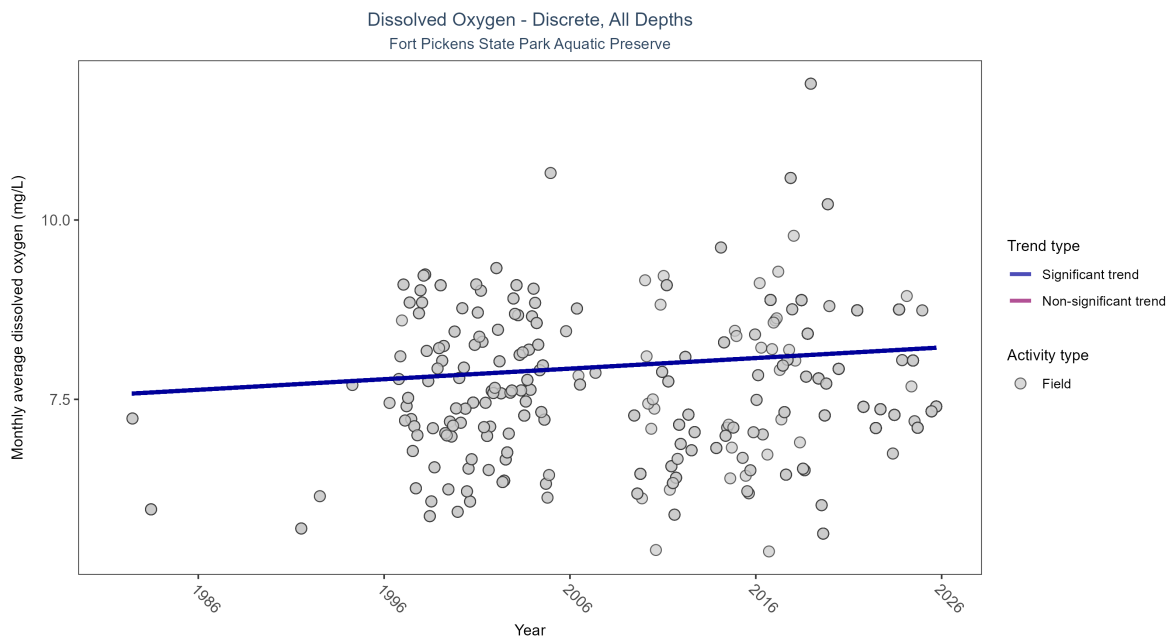


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau    | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|--------|---------------|-----------|--------|
| Field         | Significantly increasing trend | 1238         | 34              | 1982 - 2025      | 7.5                 | 0.1119 | 7.57427       | 0.01475   | 0.0193 |

Monthly average dissolved oxygen increased by 0.01 mg/L per year.



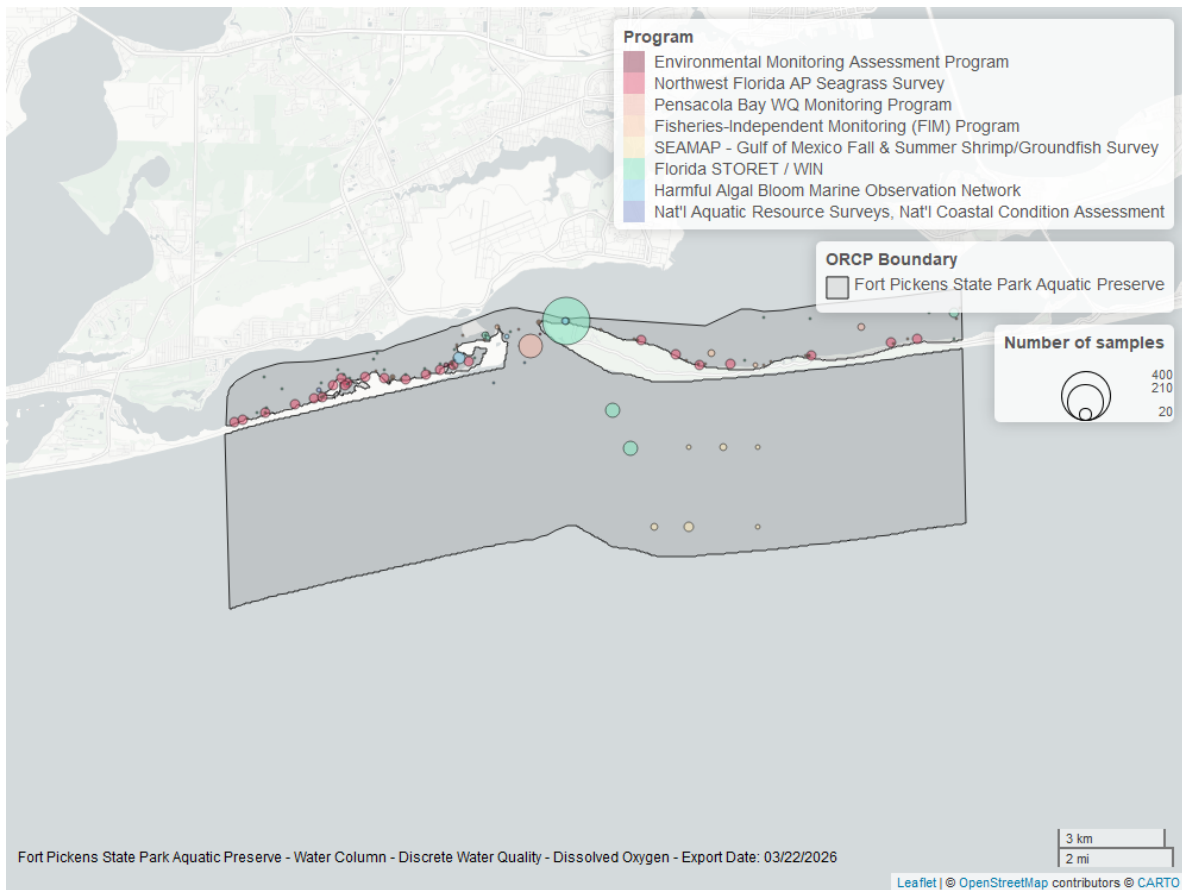


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Dissolved Oxygen Saturation - Continuous

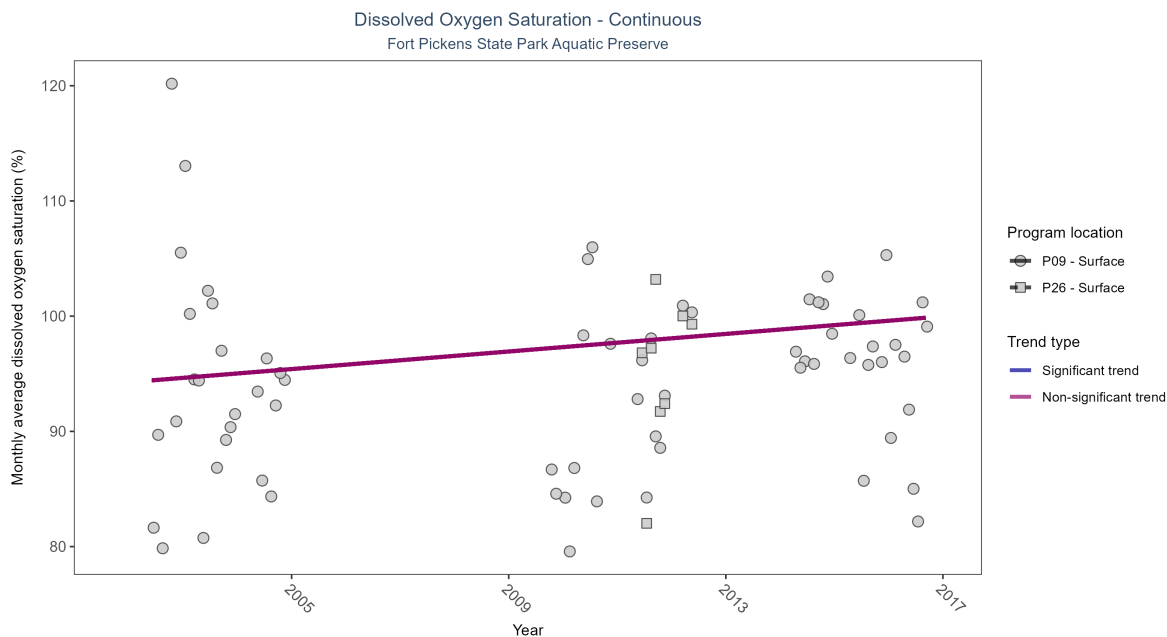


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|------|---------------|-----------|--------|
| P09              | No significant trend                 | 358          | 10              | 2002 - 2016      | 96.91               | 0.11 | 94.26         | 0.38      | 0.2446 |
| P26              | Insufficient data to calculate trend | 27           | 2               | 2011 - 2012      | 98.17               | -    | -             | -         | -      |

No detectable change in monthly average dissolved oxygen saturation was observed at one location. There was insufficient data to fit a model for one location.

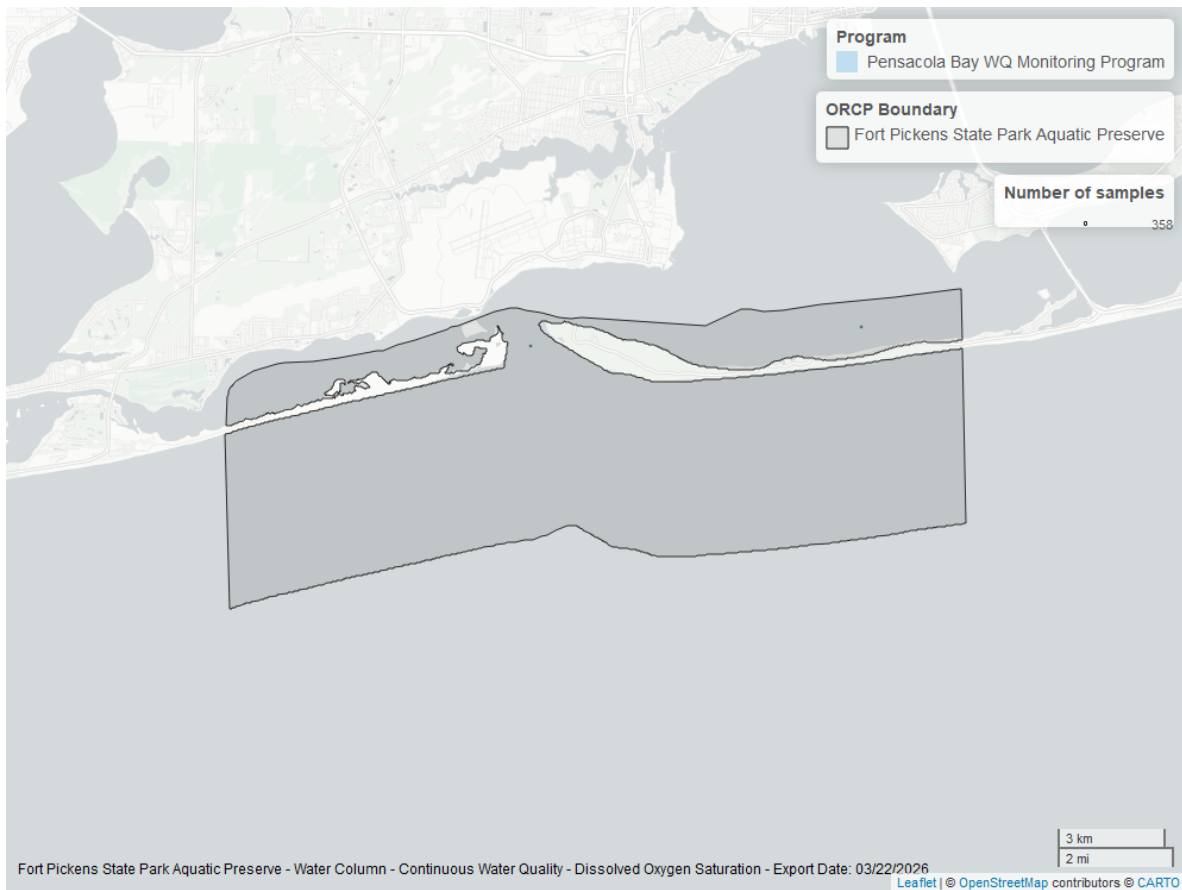


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Dissolved Oxygen Saturation - Discrete

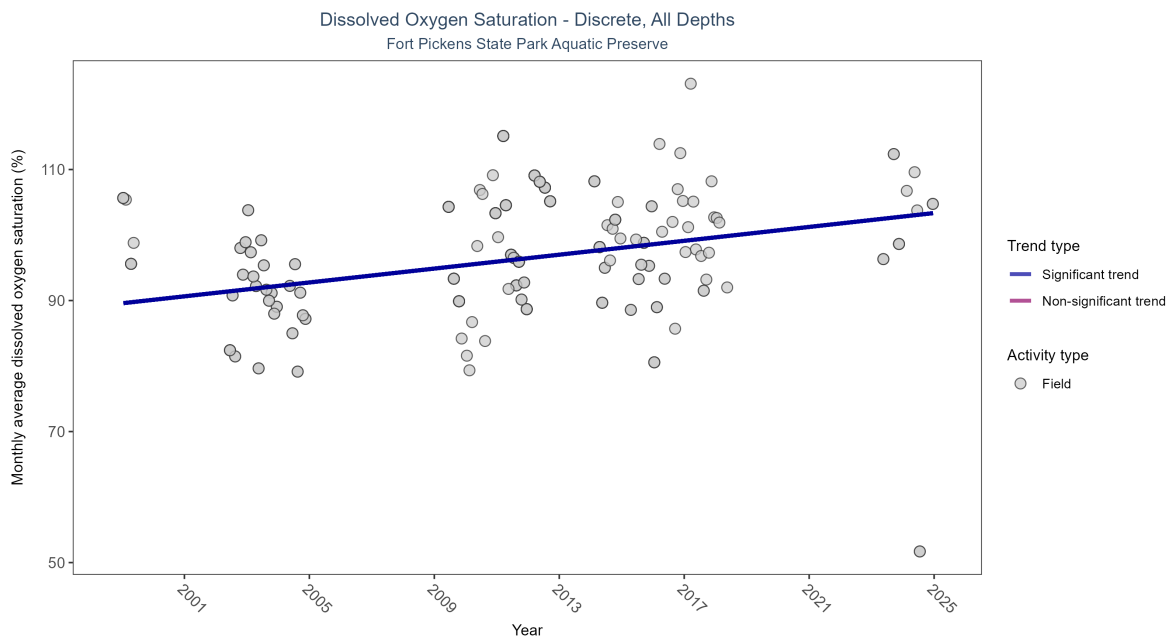


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P     |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|-------|
| Field         | Significantly increasing trend | 261          | 15              | 1999 - 2024      | 98.6                | 0.27656 | 89.58022      | 0.52972   | 5e-04 |

Monthly average dissolved oxygen saturation increased by 0.53% per year.

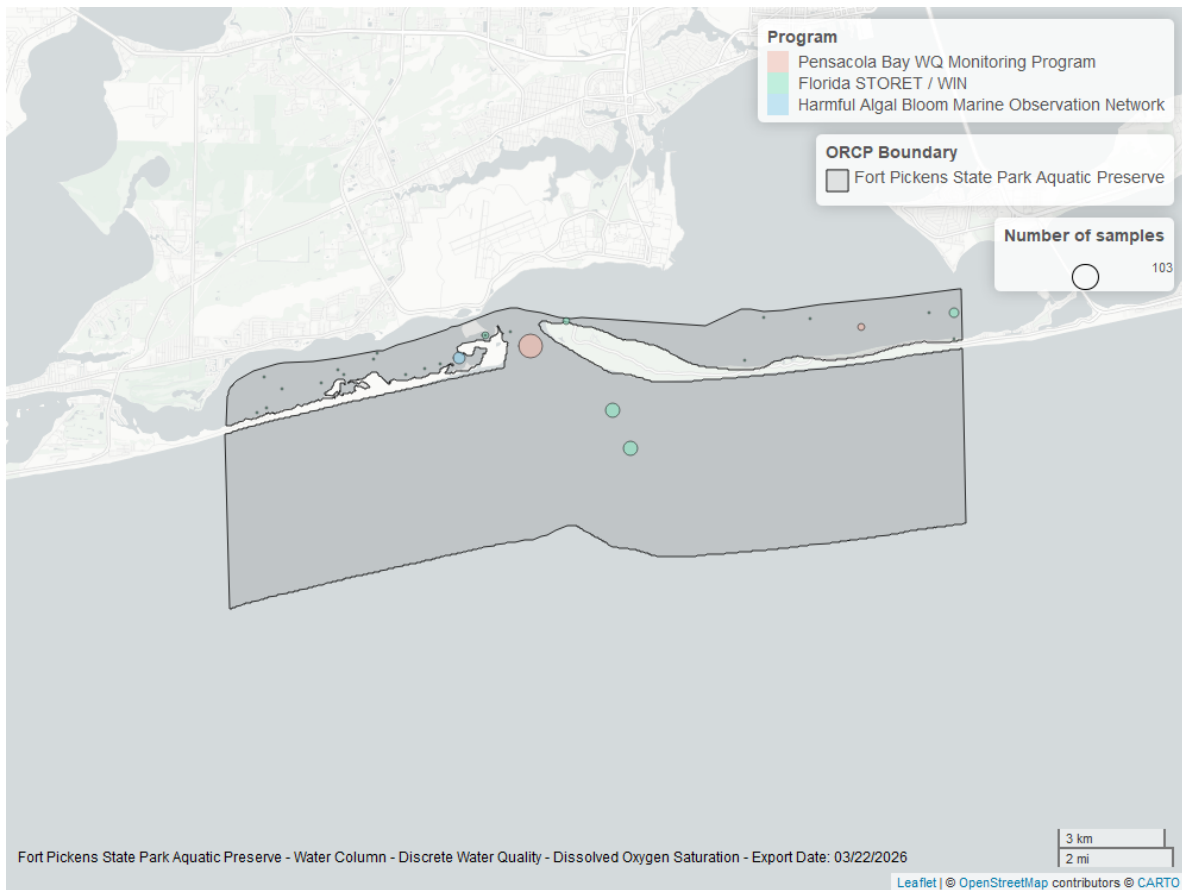


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Salinity - Discrete

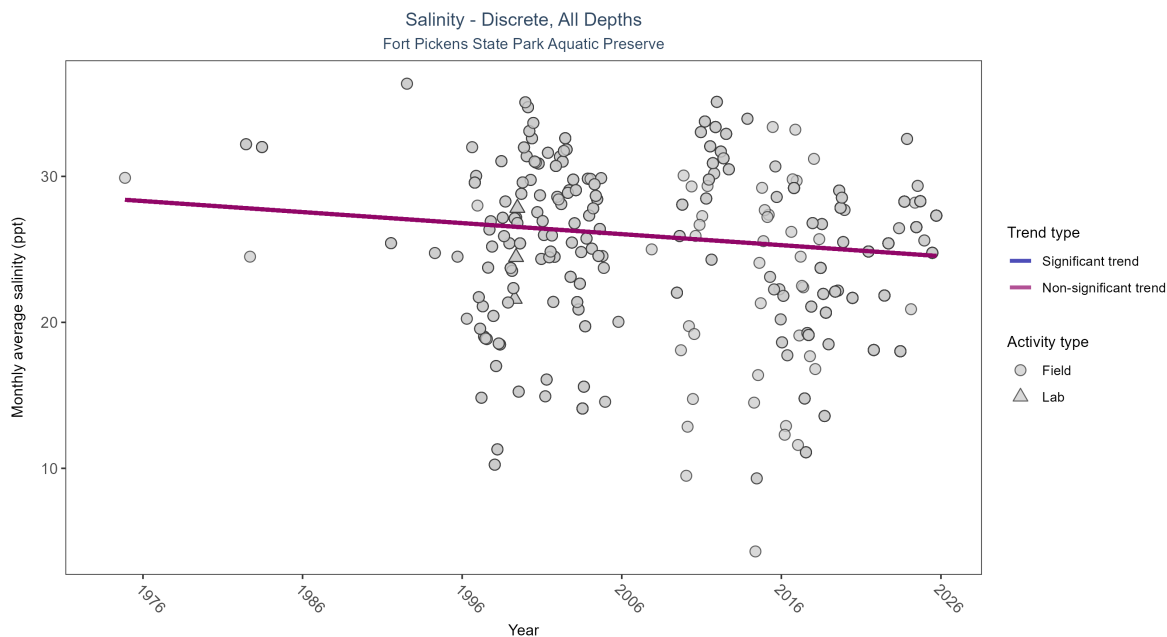


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

| Activity Type | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P      |
|---------------|----------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|--------|
| All           | No significant trend | 1242         | 35              | 1974 - 2025      | 26.3                | -0.08499 | 28.46168      | -0.07554  | 0.0787 |

Salinity showed no detectable trend between 1974 and 2025.

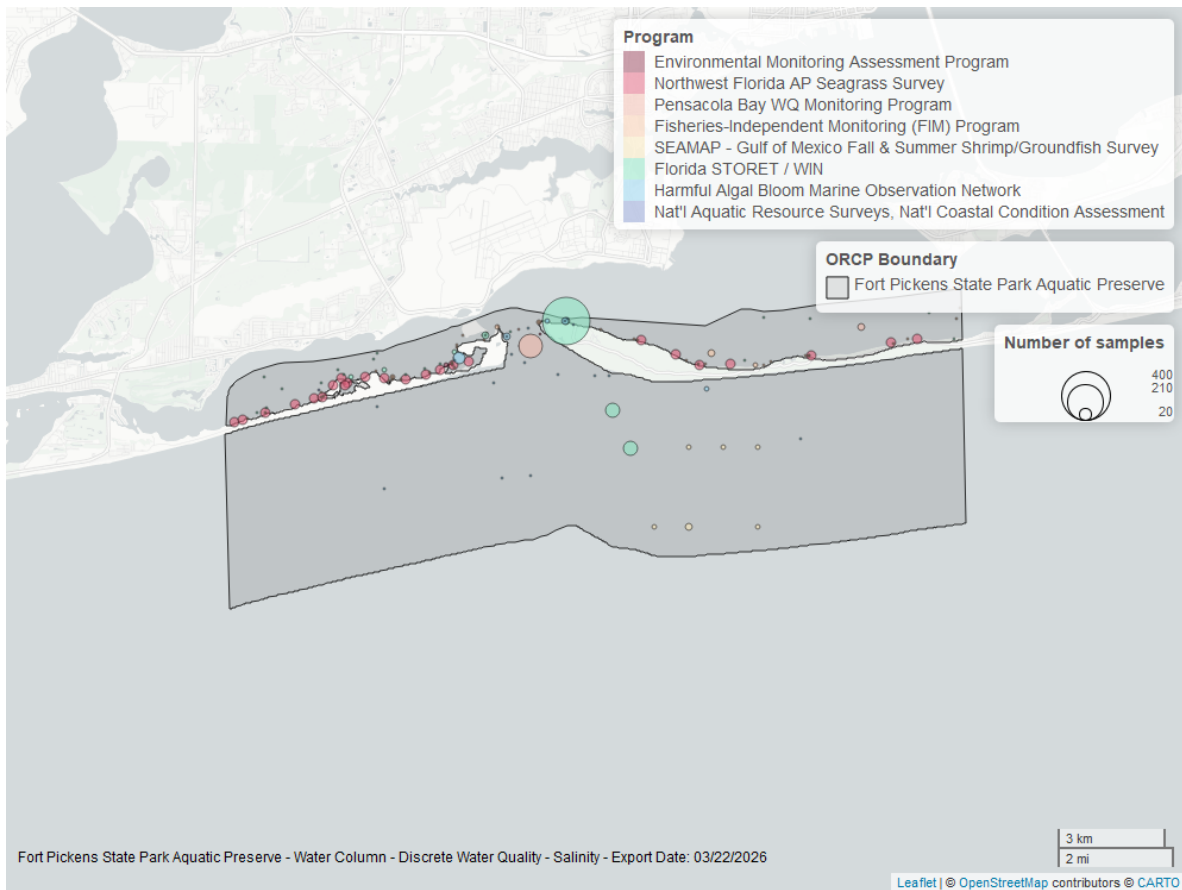


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Salinity - Continuous

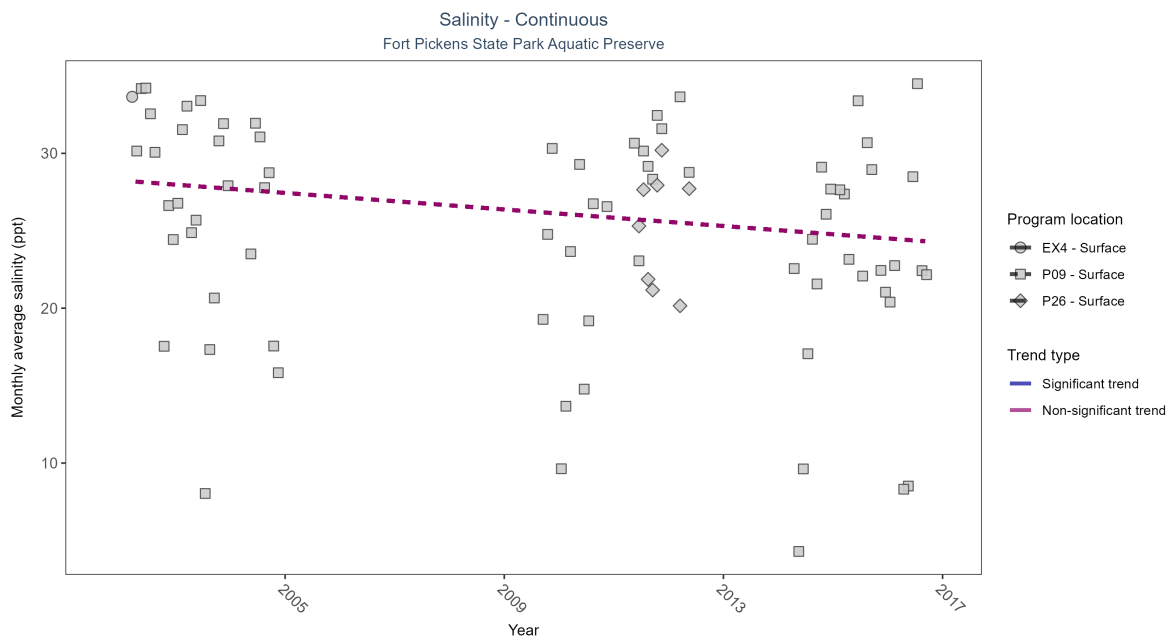


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-------|---------------|-----------|--------|
| EX4              | Insufficient data to calculate trend | 2            | 1               | 2002 - 2002      | 33.65               | -     | -             | -         | -      |
| P09              | No significant trend                 | 373          | 10              | 2002 - 2016      | 25.91               | -0.17 | 28.25         | -0.27     | 0.1296 |
| P26              | Insufficient data to calculate trend | 27           | 2               | 2011 - 2012      | 26.38               | -     | -             | -         | -      |

No detectable change in monthly average salinity was observed at one location. There was insufficient data to fit a model for two locations.



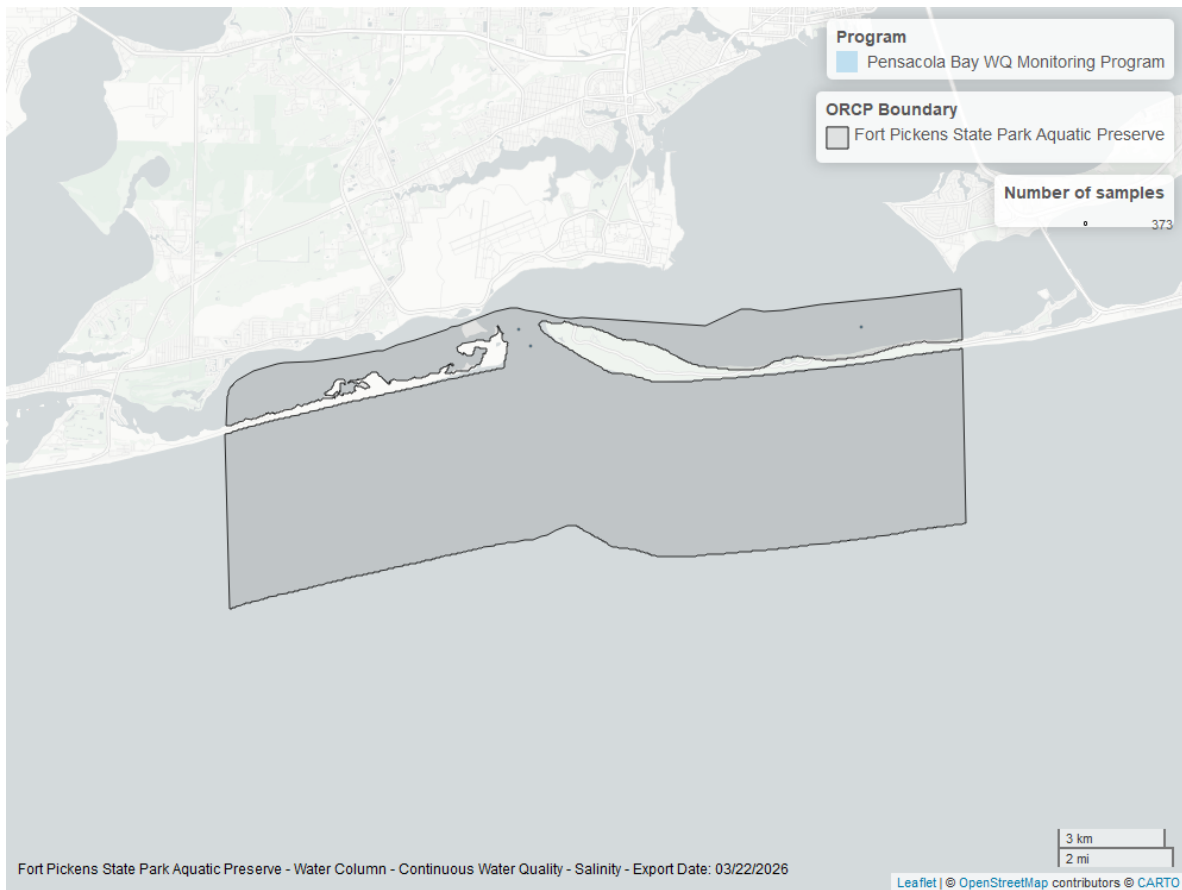


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Temperature - Continuous

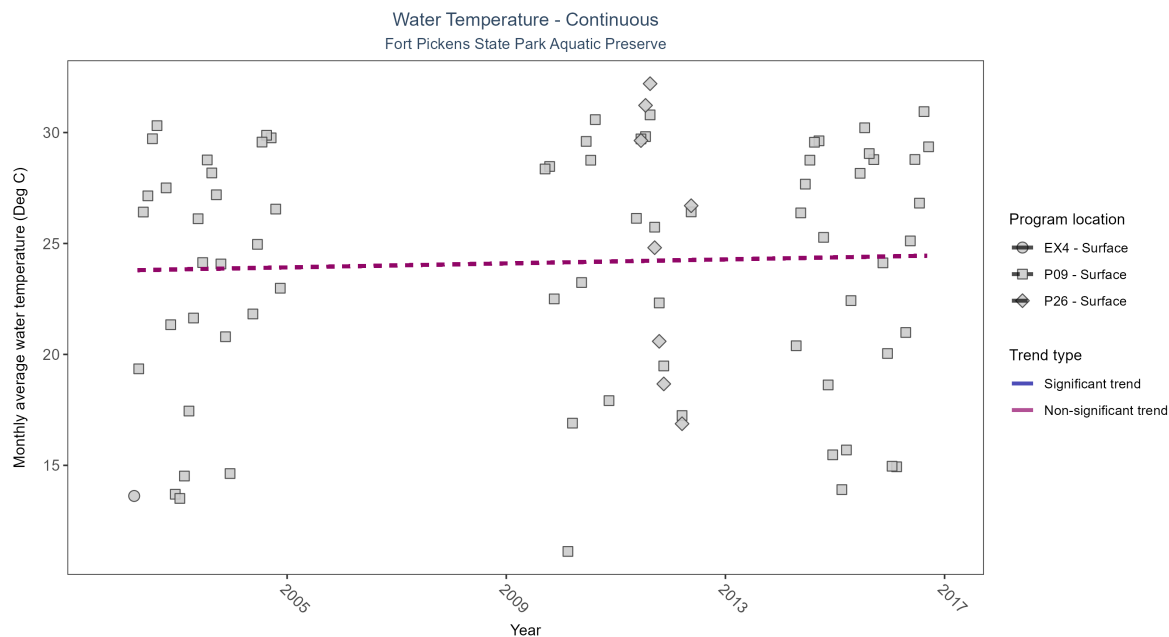


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results - Water Temperature

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-----|---------------|-----------|--------|
| EX4              | Insufficient data to calculate trend | 2            | 1               | 2002 - 2002      | 13.62               | -   | -             | -         | -      |
| P09              | No significant trend                 | 373          | 10              | 2002 - 2016      | 26.39               | 0.1 | 23.79         | 0.05      | 0.4485 |
| P26              | Insufficient data to calculate trend | 27           | 2               | 2011 - 2012      | 29.64               | -   | -             | -         | -      |

No detectable change in monthly average water temperature was observed at one location. There was insufficient data to fit a model for two locations.

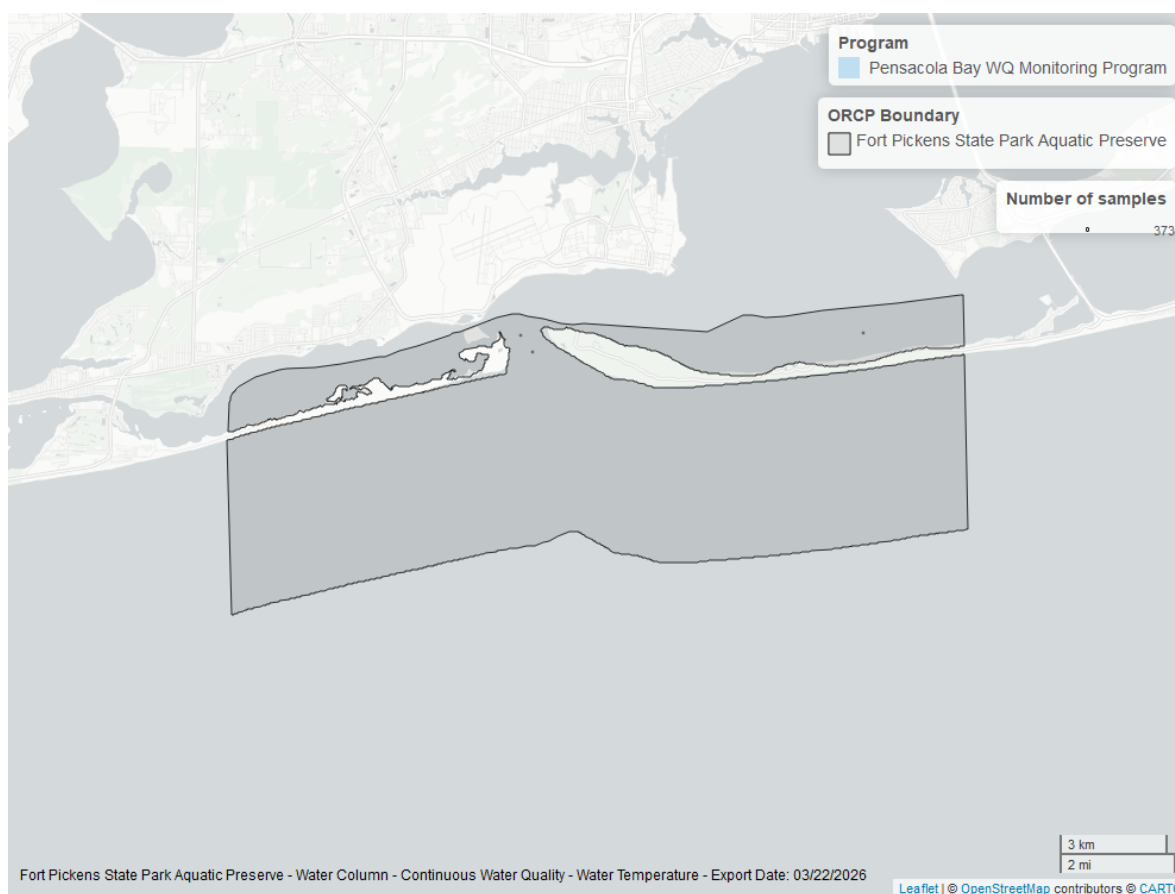


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Temperature - Discrete

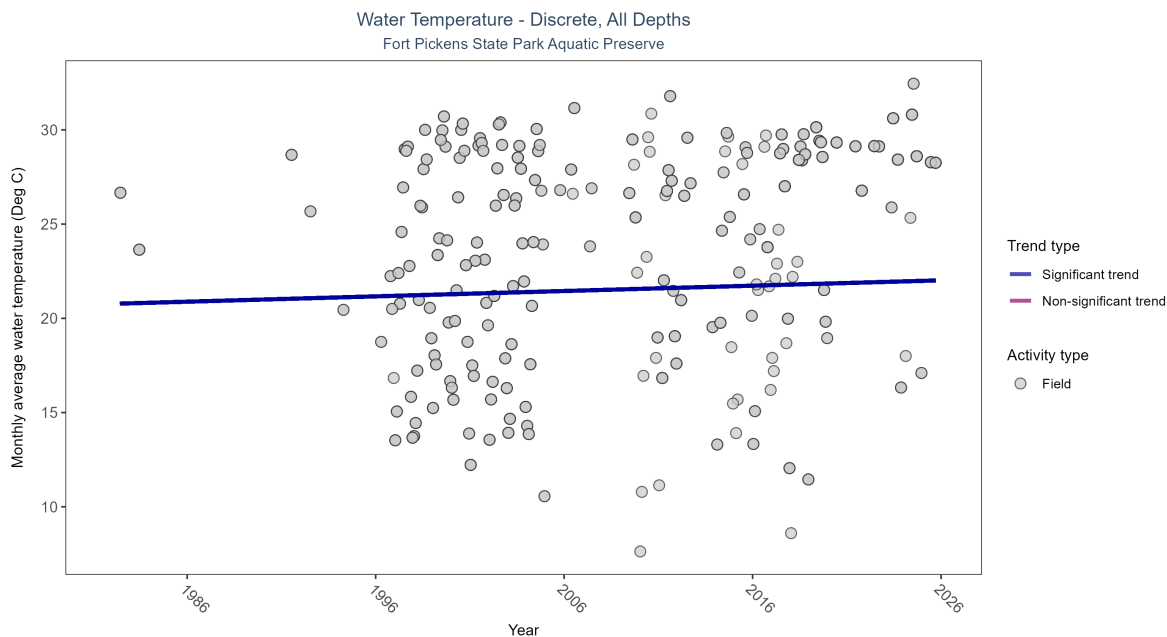


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Water Temperature

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau    | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|--------|---------------|-----------|--------|
| Field         | Significantly increasing trend | 1249         | 34              | 1982 - 2025      | 27.3                | 0.0831 | 20.76838      | 0.02835   | 0.0492 |

Monthly average water temperature increased by 0.03°C per year.

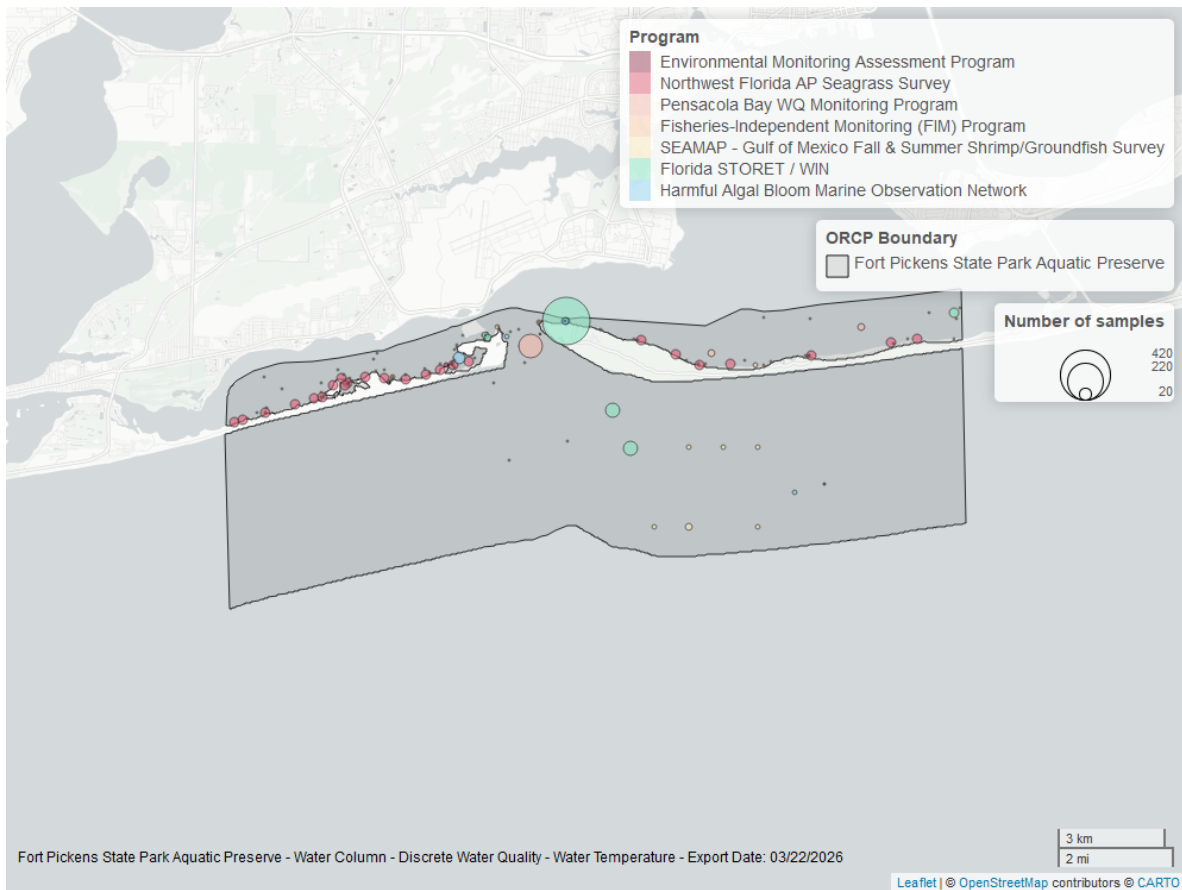


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## pH - Continuous

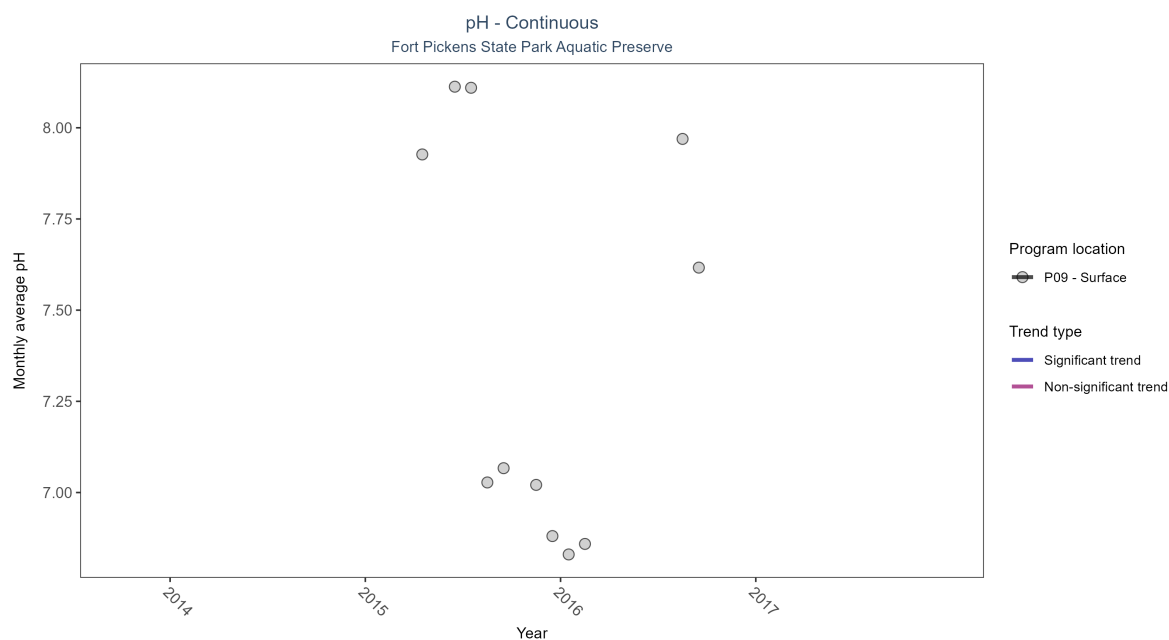


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results - pH

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-----|---------------|-----------|---|
| P09              | Insufficient data to calculate trend | 157          | 2               | 2015 - 2016      | 7.06                | -   | -             | -         | - |

There was insufficient data to fit a model for one location.

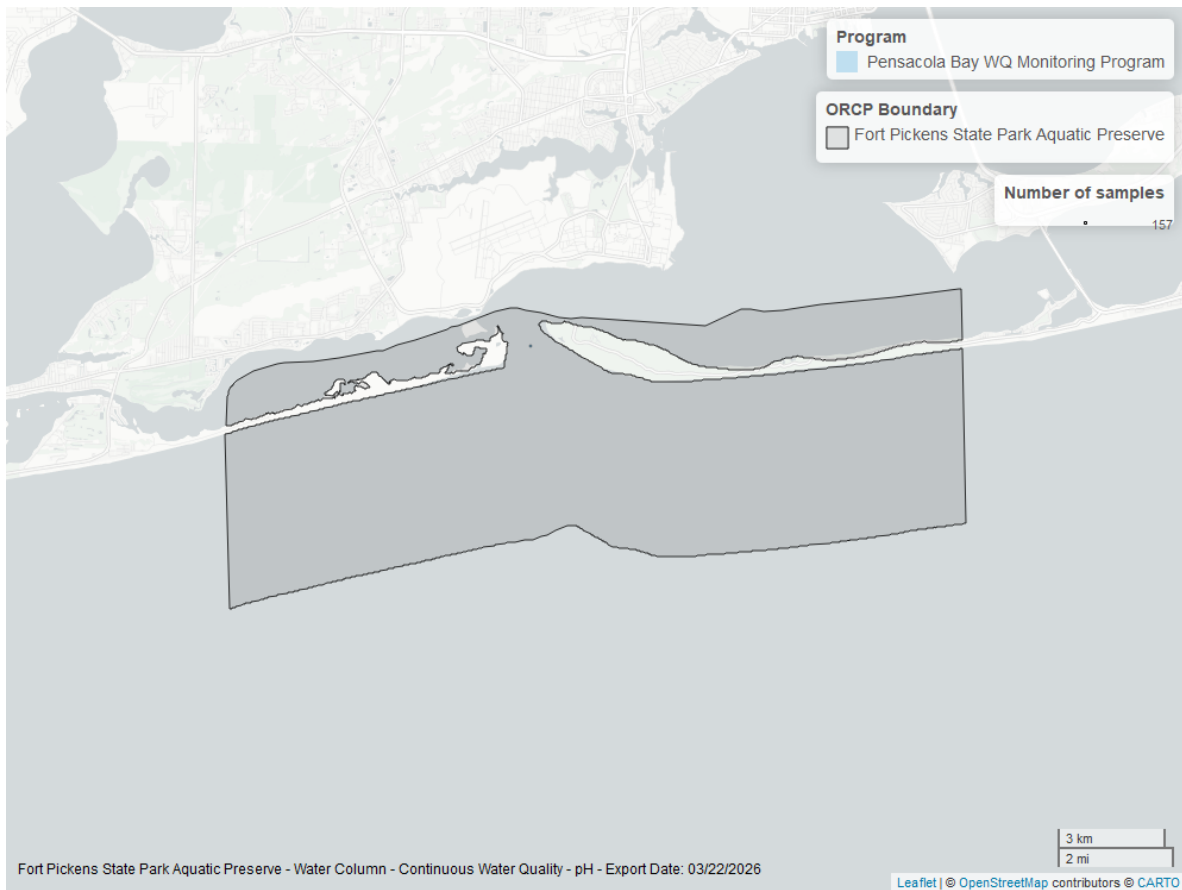


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## pH - Discrete

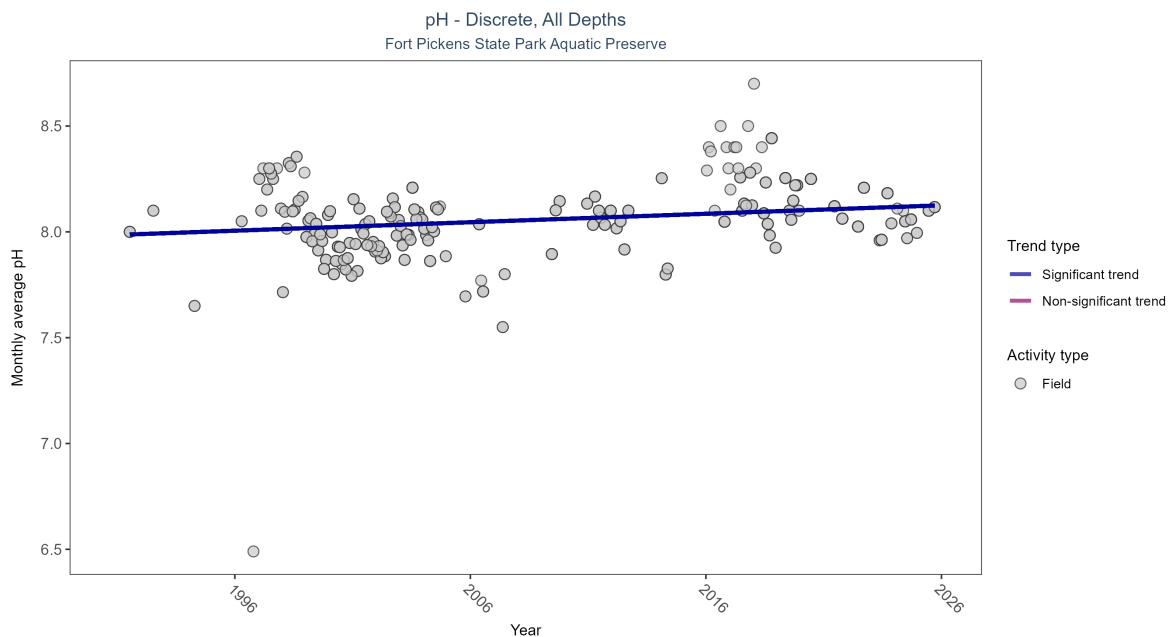


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - pH

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| Field         | Significantly increasing trend | 1003         | 30              | 1991 - 2025      | 8.1                 | 0.17298 | 7.98539       | 0.004     | 0.0077 |

Monthly average pH increased by less than 0.01 pH units per year.



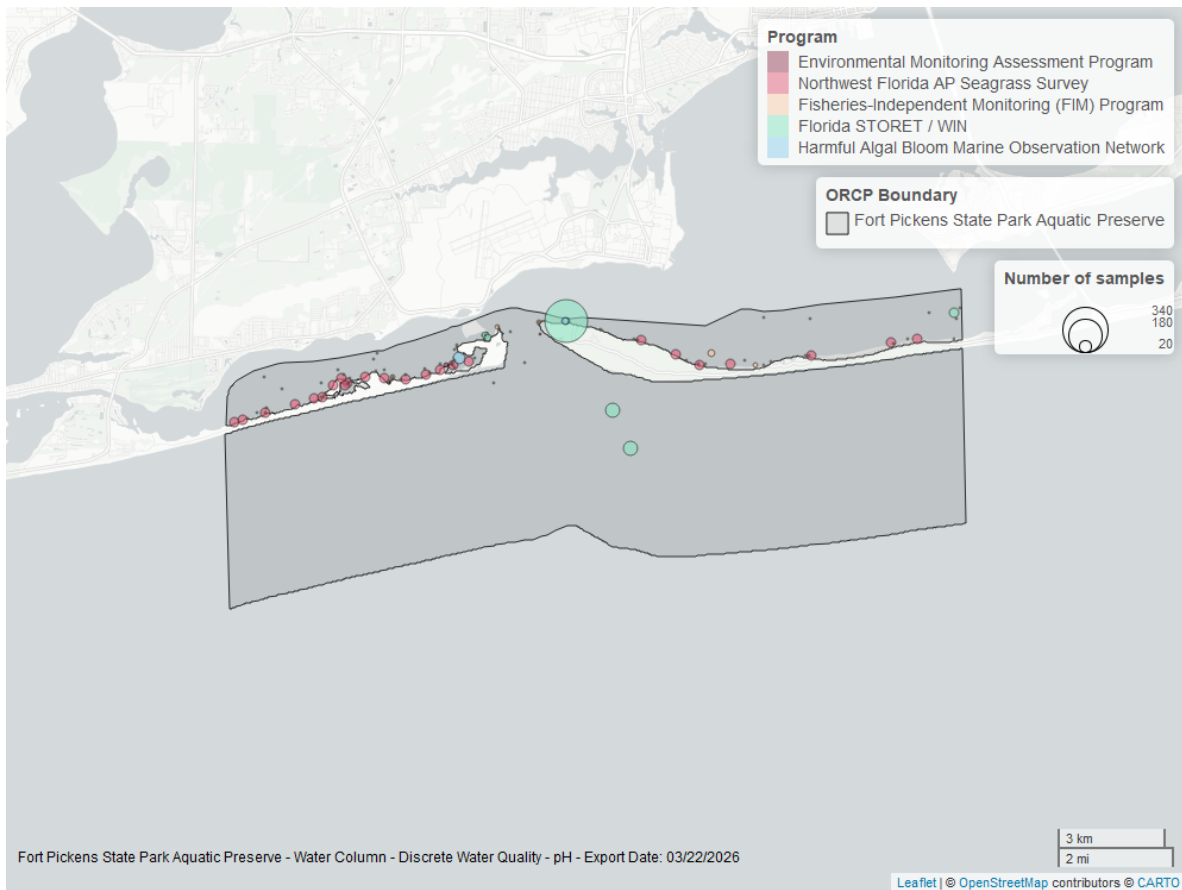


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Clarity

### Turbidity - Continuous

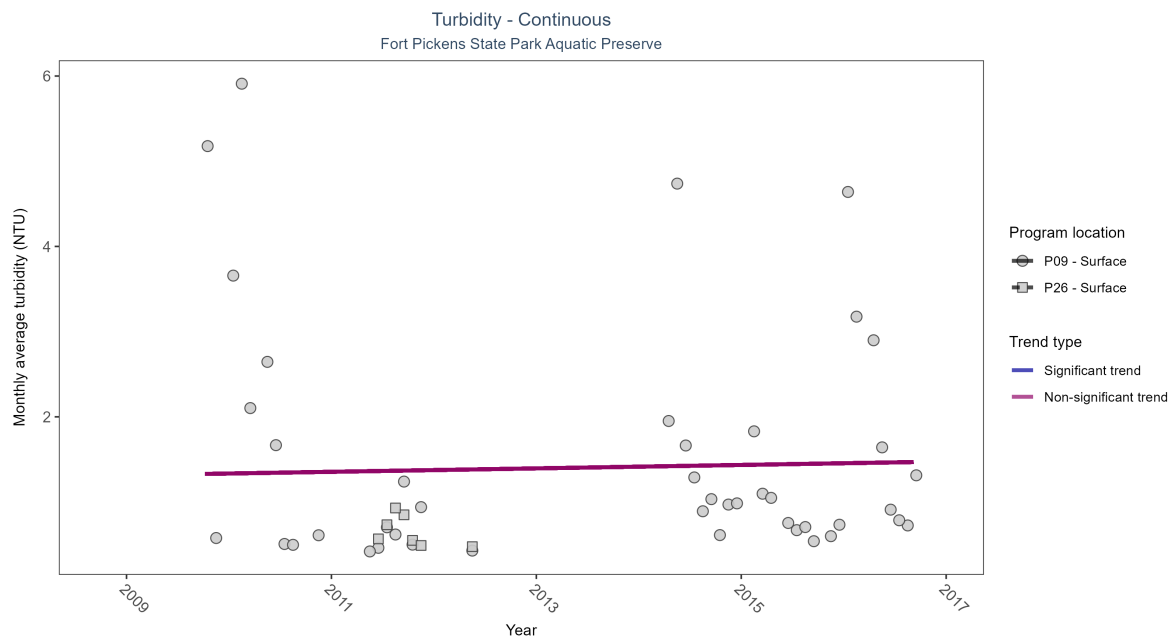


Figure 25: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: Seasonal Kendall-Tau Results - Turbidity

| Program Location | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P     |
|------------------|--------------------------------------|--------------|-----------------|------------------|---------------------|------|---------------|-----------|-------|
| P09              | No significant trend                 | 275          | 7               | 2009 - 2016      | 0.85                | 0.07 | 1.31          | 0.02      | 0.382 |
| P26              | Insufficient data to calculate trend | 25           | 2               | 2011 - 2012      | 0.62                | -    | -             | -         | -     |

No detectable change in monthly average turbidity was observed at one location. There was insufficient data to fit a model for one location.

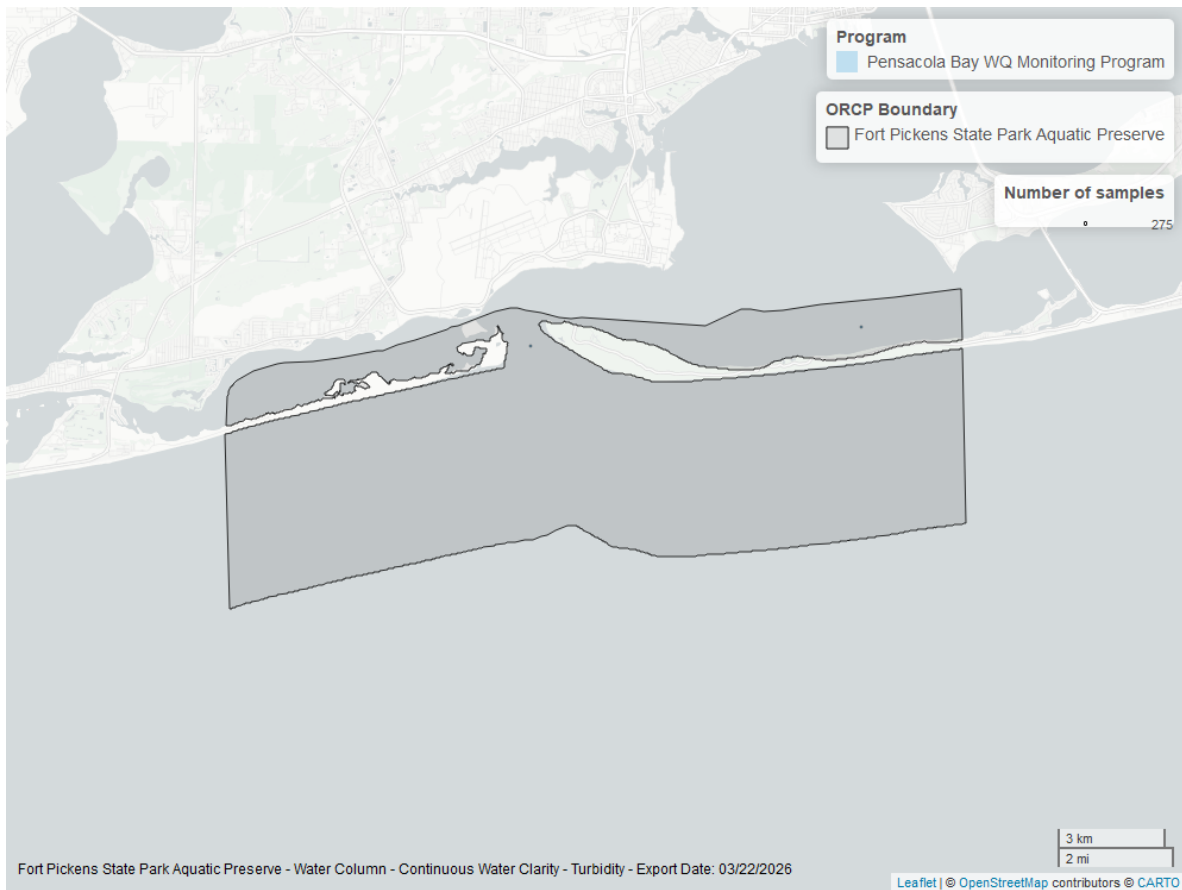


Figure 26: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Turbidity - Discrete

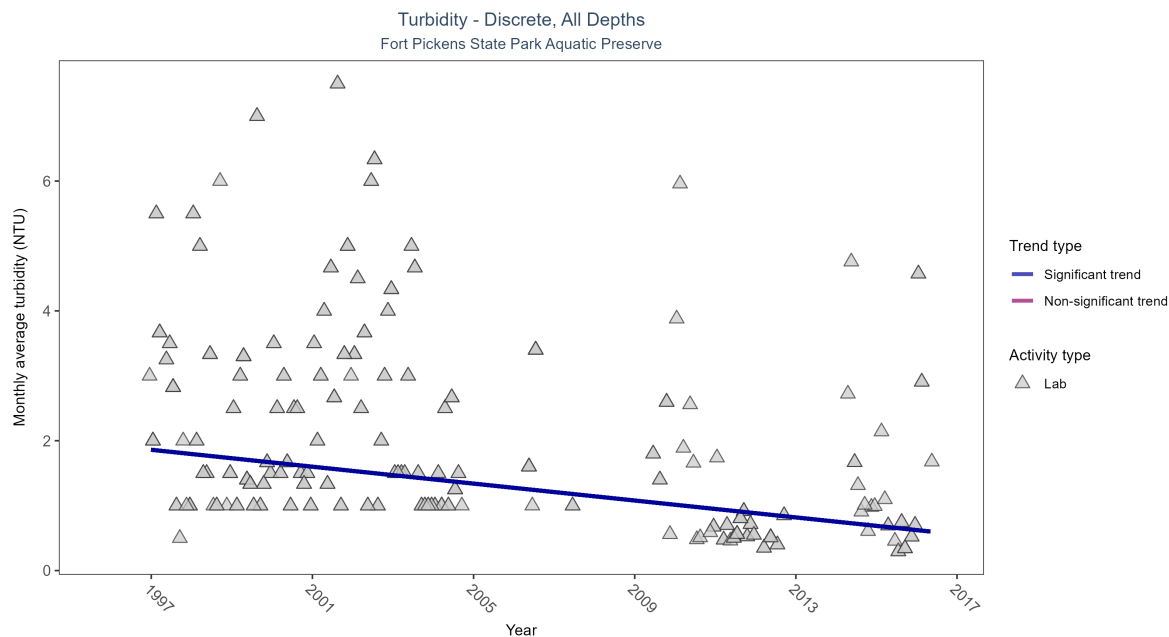


Figure 27: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 14: Seasonal Kendall-Tau Results for - Turbidity

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|---|
| Lab           | Significantly decreasing trend | 313          | 18              | 1996 - 2016      | 1.0122              | -0.31425 | 1.92558       | -0.06511  | 0 |

Monthly average turbidity decreased by 0.07 NTU per year, indicating an increase in water clarity.

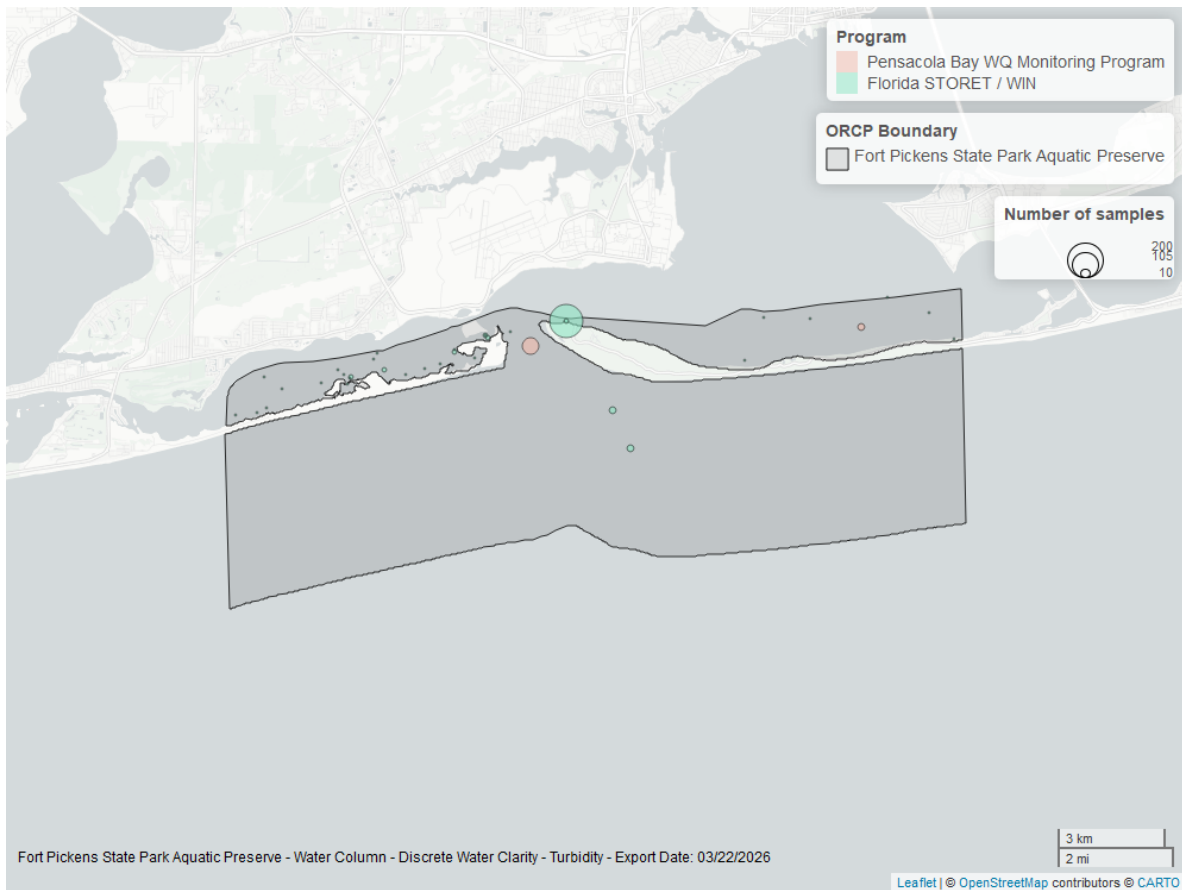


Figure 28: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Total Suspended Solids - Discrete

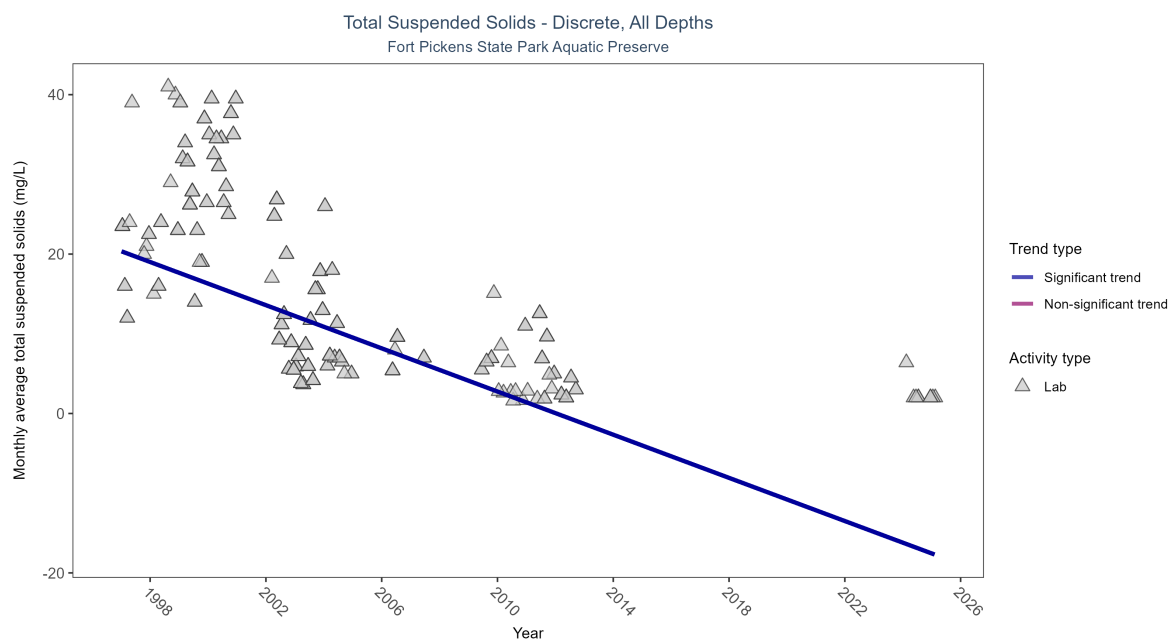


Figure 29: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Total Suspended Solids

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|---|
| Lab           | Significantly decreasing trend | 264          | 15              | 1997 - 2025      | 9.2                 | -0.63671 | 20.36003      | -1.35417  | 0 |

Monthly average total suspended solids decreased by 1.35 mg/L per year, indicating an increase in water clarity.

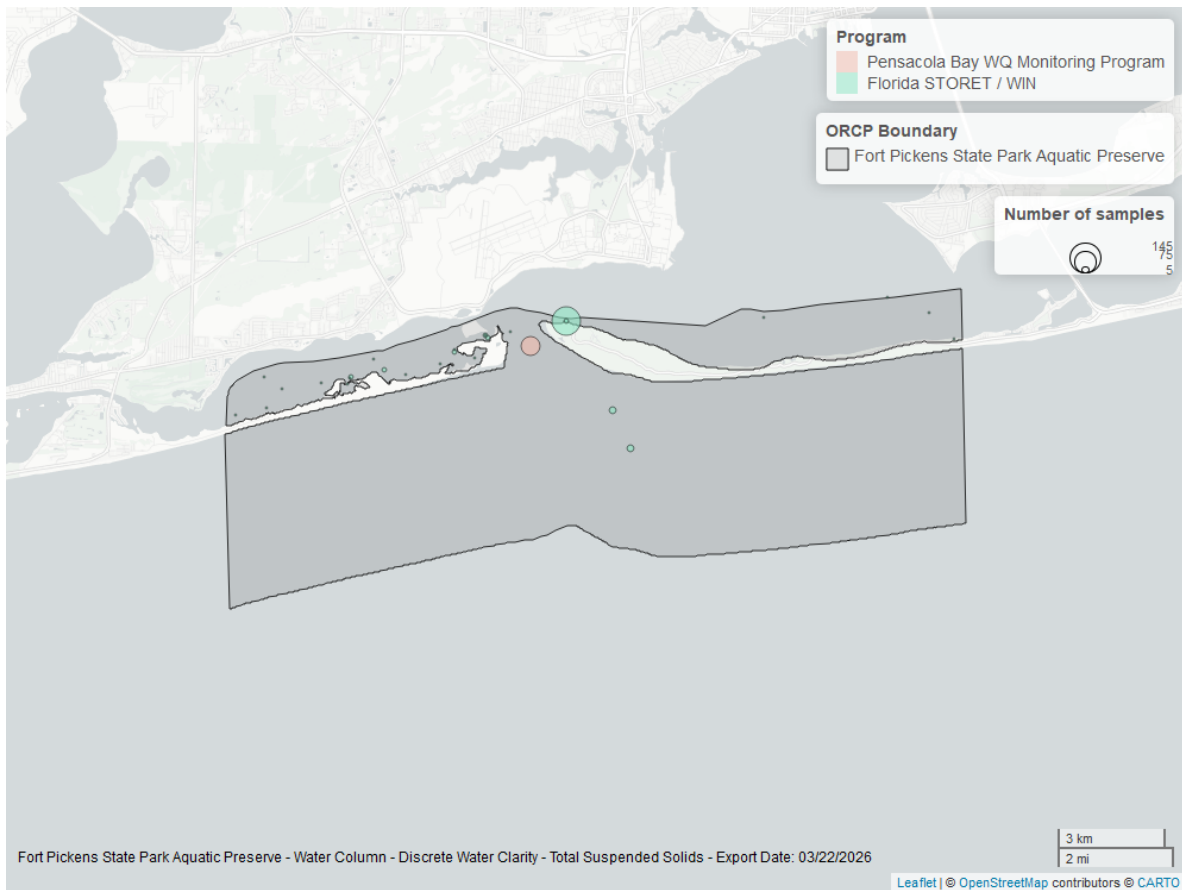


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Chlorophyll a, Uncorrected for Pheophytin - Discrete

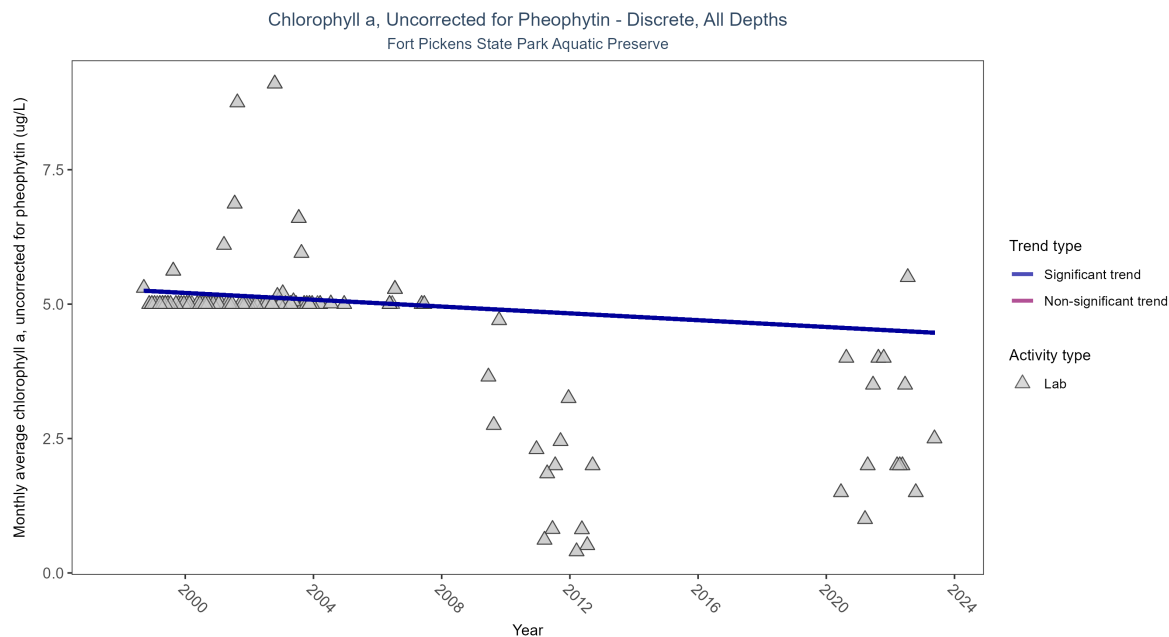


Figure 31: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|---|
| Lab           | Significantly decreasing trend | 224          | 17              | 1998 - 2023      | 5                   | -0.32141 | 5.27153       | -0.03167  | 0 |

Monthly average chlorophyll a, uncorrected for pheophytin, decreased by 0.03 µg/L per year, indicating an increase in water clarity.



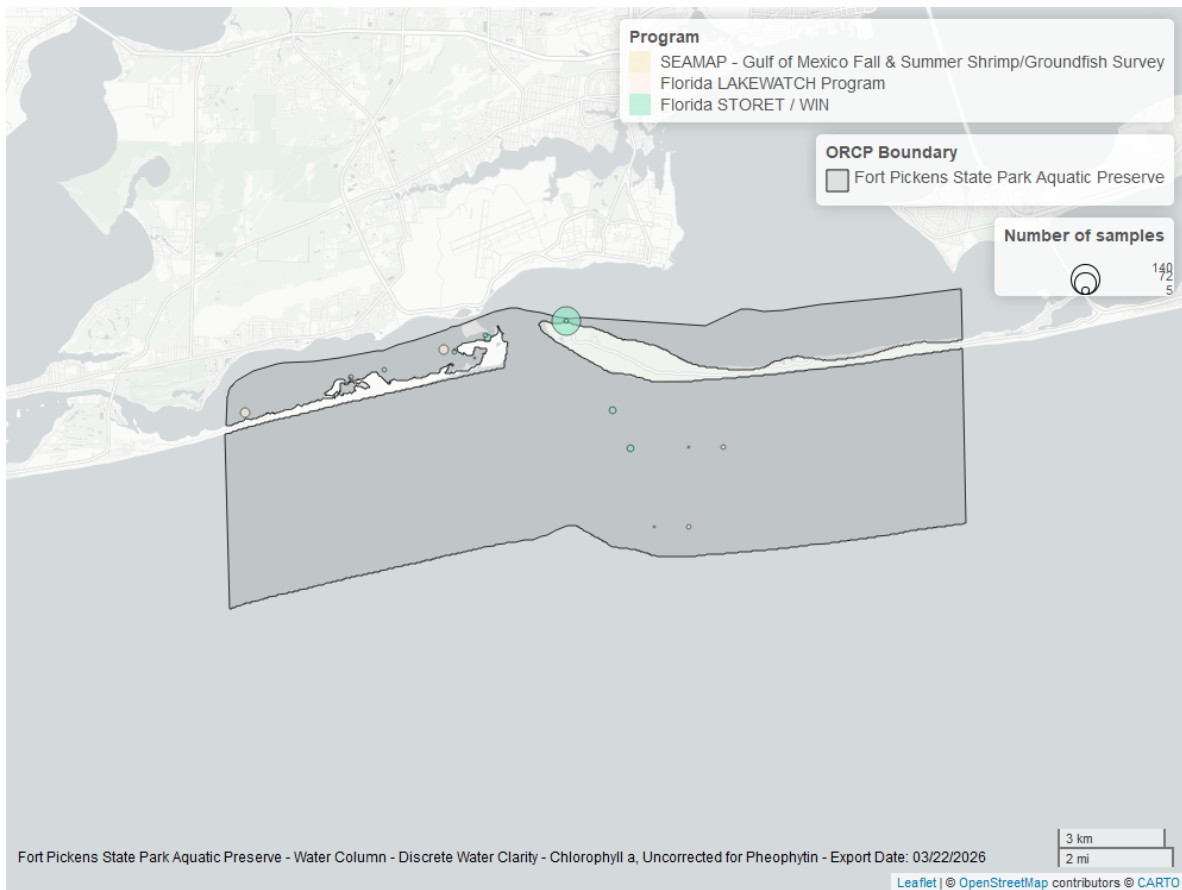


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Chlorophyll a, Corrected for Pheophytin - Discrete

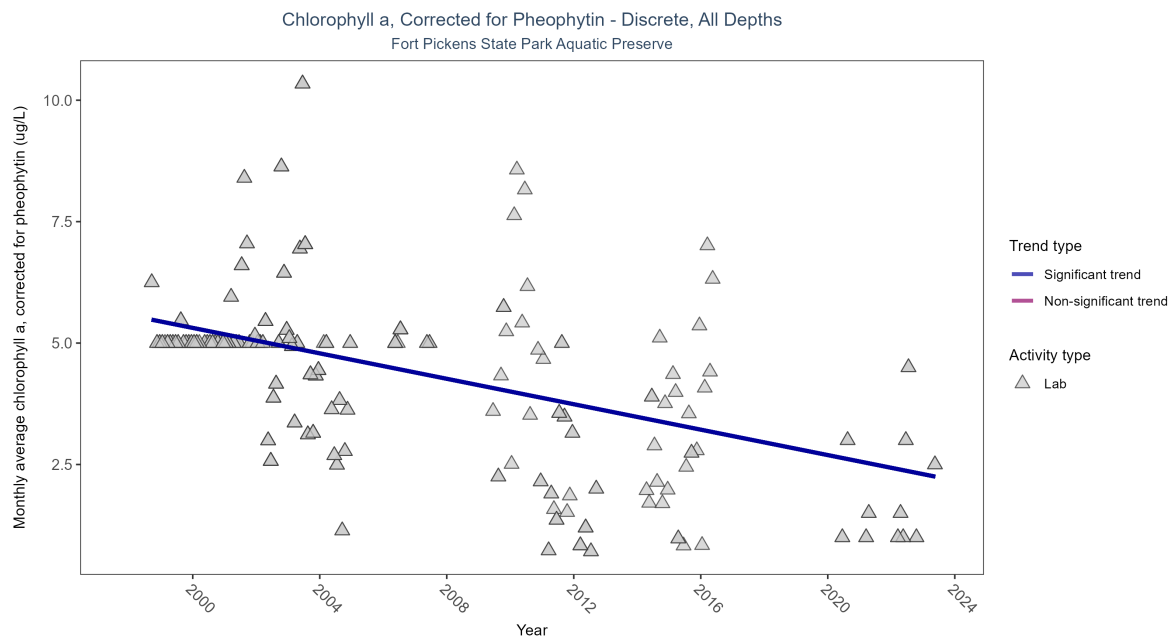


Figure 33: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|---|
| Lab           | Significantly decreasing trend | 307          | 20              | 1998 - 2023      | 5                   | -0.3836 | 5.57382       | -0.13091  | 0 |

Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.13  $\mu\text{g/L}$  per year, indicating an increase in water clarity.

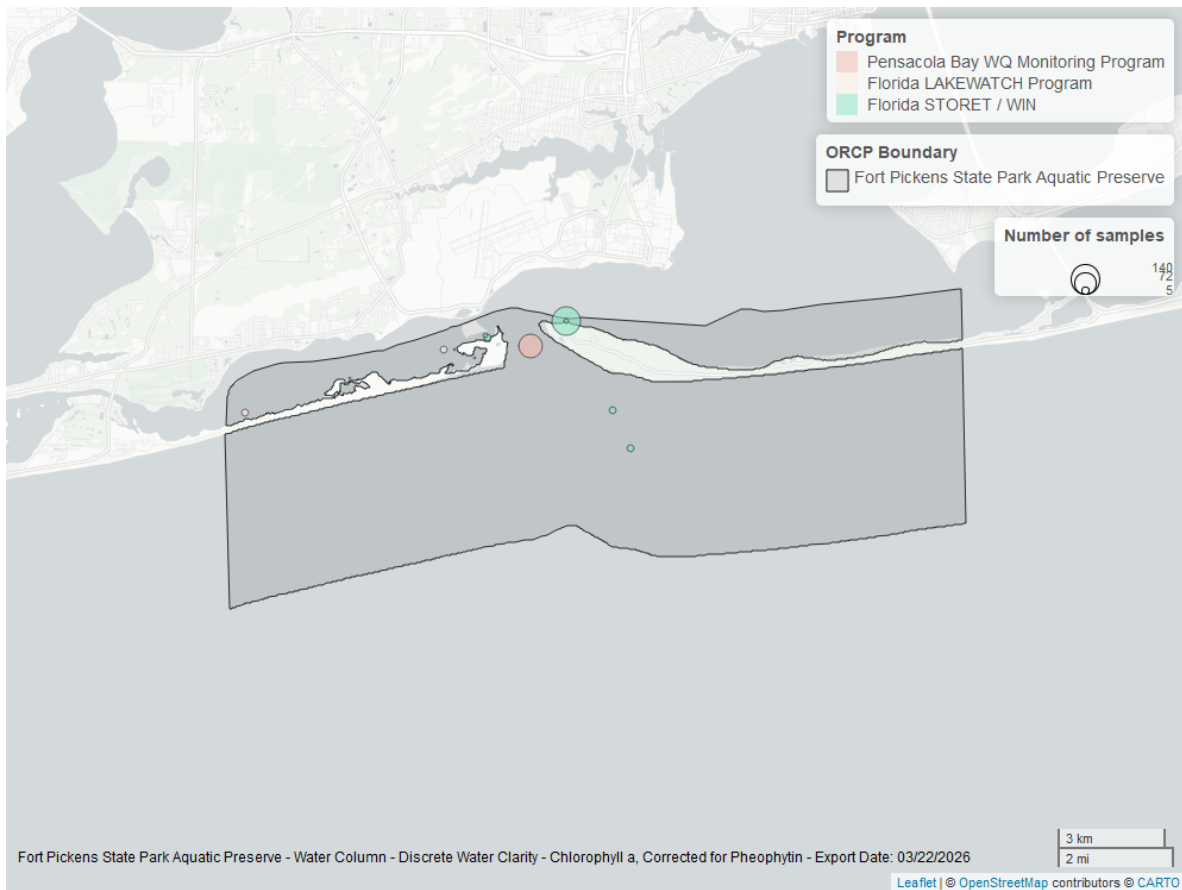


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Secchi Depth - Discrete

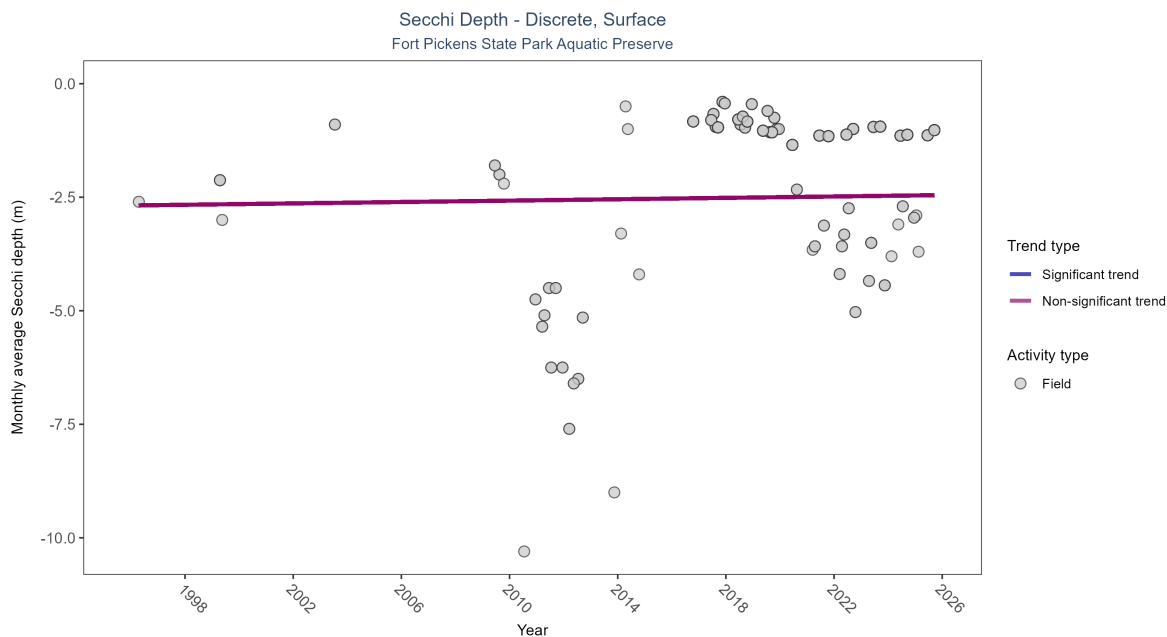


Figure 35: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 18: Seasonal Kendall-Tau Results for - Secchi Depth

| Activity Type | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|----------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| Field         | No significant trend | 572          | 19              | 1996 - 2025      | -0.91441            | 0.00486 | -2.68182      | 0.00762   | 0.7251 |

Secchi depth showed no detectable trend between 1991 and 2025.

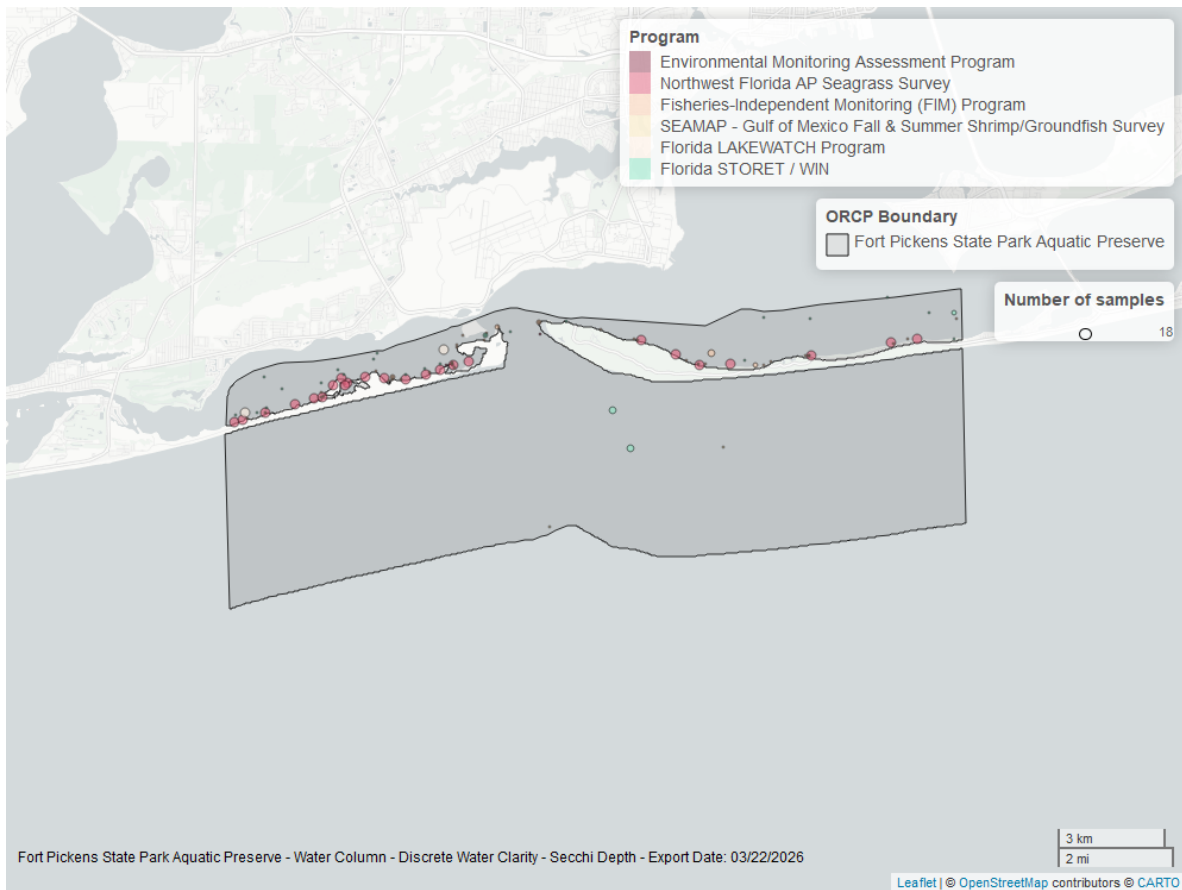


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

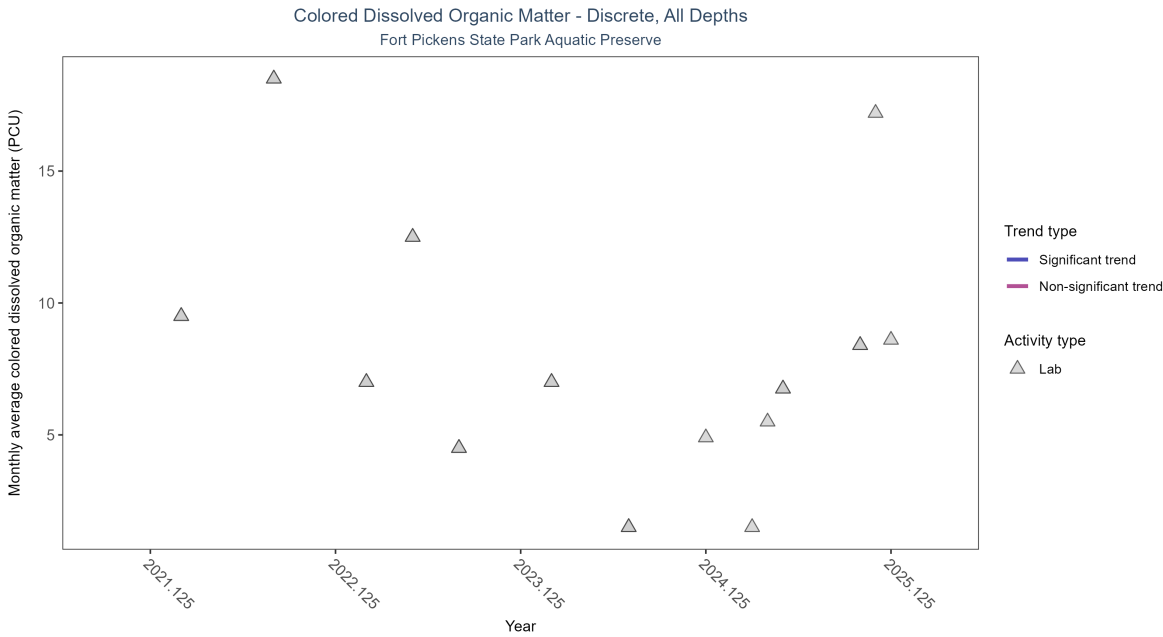


Figure 37: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 19: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

| Activity Type | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-----|---------------|-----------|---|
| Lab           | Insufficient data to calculate trend | 23           | 5               | 2021 - 2025      | 7                   | -   | -             | -         | - |

There was insufficient data to fit a model for colored dissolved organic matter.

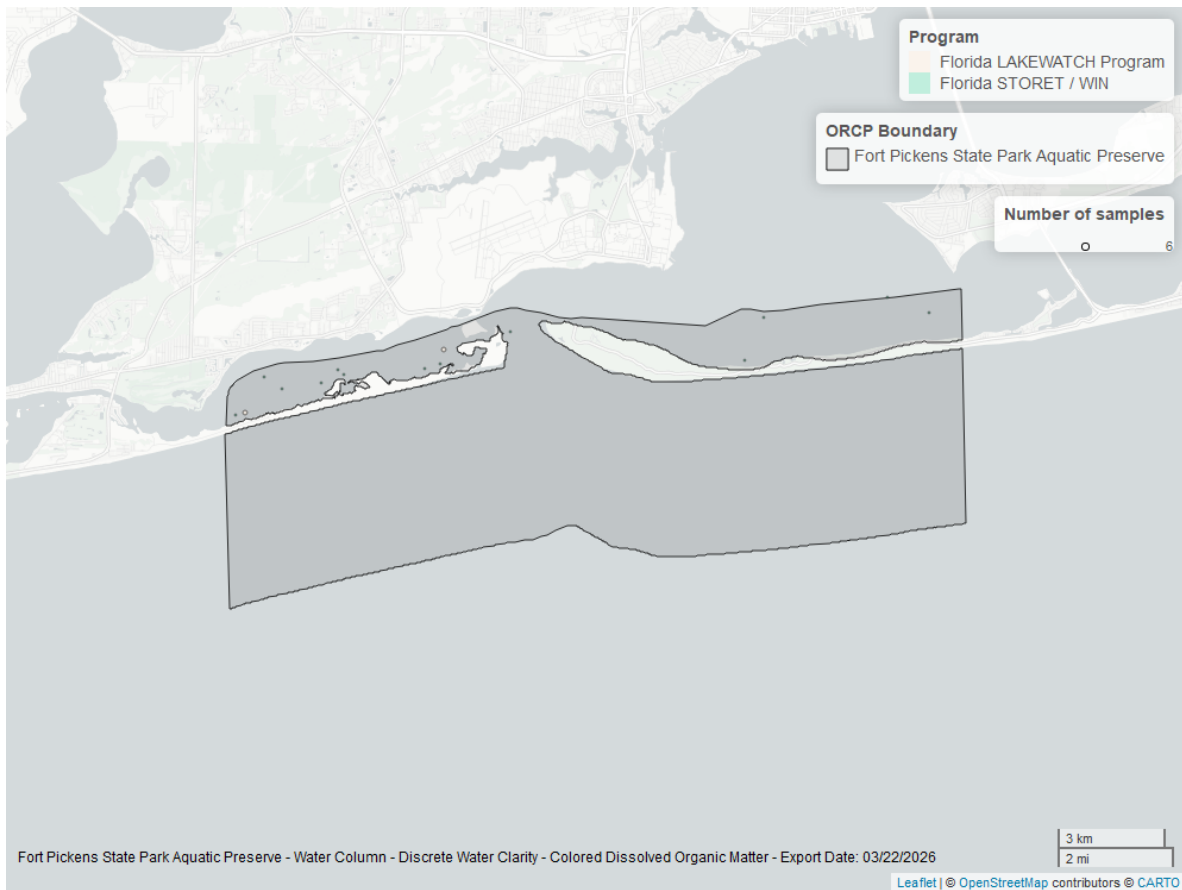


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.